

MATH SURVEY 1998

Volume LXXII • No. 1 • Stuyvesant High School • Department of Mathematics • New York, NY

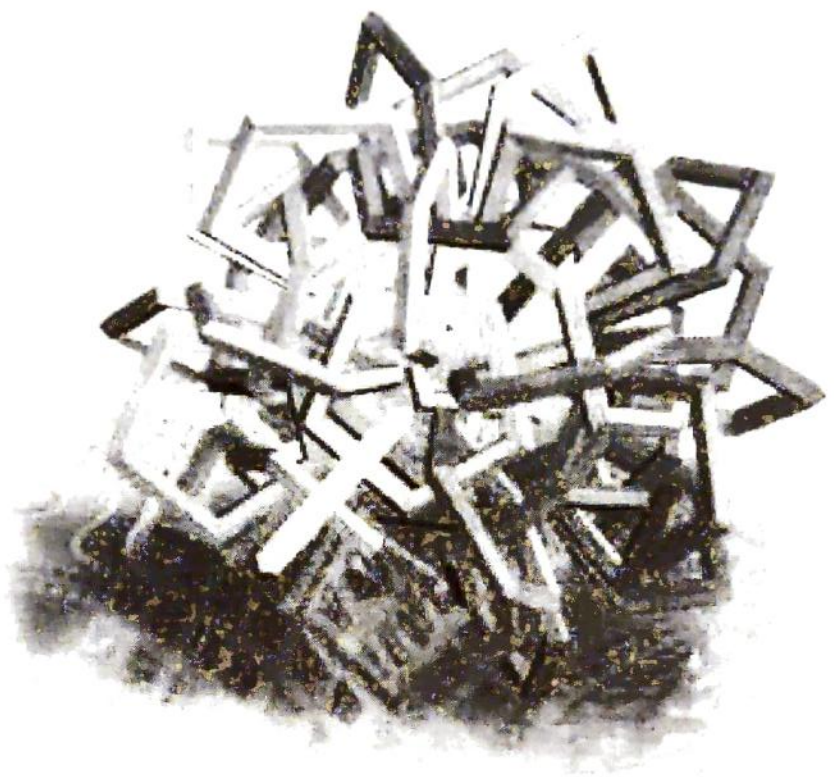


Table of Contents

A Necessary and Sufficient Condition for an n -lattice to be Hamiltonian, <i>Alex Tang</i>	5
Periodic Properties of the Fibonacci Sequence Modulo m , <i>James Chen</i>	9
Minkowski's Theorem, <i>Chia-Hao La</i>	14
Genetics with Set Theory, <i>Edmund Chou</i>	16
Some Remarkable Inequalities, <i>Regina Eum</i>	19
Chaos Theory and Newton's Method, <i>Nelson Uhan</i>	21
The Pigeonhole Principle, <i>Weikai Lang</i>	23
Fermat's Last Theorem: The Basic Idea to the Solution, <i>Zardosht Kasheff</i>	25
Continued Fractions, <i>Jae Jang</i>	28
Options Pricing Using the Black-Scholes Model, <i>John Choi</i>	32
Special Magic Squares, <i>Sung Hoon Kim</i>	35
An Interview with Daniel Stronger, <i>Boris Granovskiy</i>	37
AP Calculus Review.....	40
An Ellipse Poem, <i>Ezra Oberfield</i>	

Math Survey

Volume LXXII Number 1

Spring 1998

The Mathematics Publication of Stuyvesant High School



Department of Mathematics

Stuyvesant High School

345 Chambers Street

New York, NY 10282

212-312-4800

Principal

Ms. Jinx Cozzi Perullo

Department Chairman

Mr. Stuart Weinberg

Faculty Advisor

Mr. Allen Clancy

Editor-in-Chief

Edmund Chou

Associate Editors

Regina Eum and Weikai Lang

Math Editors

Jae Jang

Zardosht Kasheff

Math Staff

Arnob Bhattacharya

Amy Yang

Literary Editor

Sung Hoon Kim

Literary Staff

James Chen

Tony Lin

Kunihiko Mikuriya

Soojin Yim

Business Manager

Victoria Hsu

Business Staff

Eric Chen

Cindy Lin

Tiffany Lin

Art and Puzzles Staff

Jane Chen

Brandon Irizarry

Ezra Oberfield

Sandy Sheng

Layout Editors

John Choi

Nelson Uhan

Contributors

Aaron Goldsmid

Boris Granovskiy

Chia-Hao La

Alex Tang

Cover Design Edmund Chou and Nelson Uhan; Graphic by Wolfram Research, Inc. (<http://www.wri.com>)

From the Editor-in-Chief

This is the seventy-second volume of *Math Survey*, the official mathematics publication of Stuyvesant High School. Unfortunately *Math Survey* was not able to continue its three year tradition of publishing two issues, which I found extremely difficult to maintain. I take full responsibility and apologize to those who were anticipating a second issue.

As Editor-in-Chief, I would first like to thank the *Math Survey* staff for their hard work and cooperation throughout the year. Many of our staff members graduated last year, and thus our returning staff members were essential to our publication. In addition to the returning members, many new staff members proved themselves to be equally capable. I would also like to extend my gratitude to all the mathematics teachers in Stuyvesant High School, for I have recently learned to appreciate their guidance. They have all taught the members of our staff, who in turn were able to contribute to publishing this edition of *Math Survey*. Furthermore, I would like to thank Mr. Allen Clancy, our faculty advisor, for his helpful feedback, especially concerning a new cover design. Finally, I would like to acknowledge the support from all the advertisers, sponsors, and readers of *Math Survey*.

To all who make *Math Survey* possible, I hope that you will enjoy this issue.

Sincerely,

A handwritten signature in black ink, appearing to read 'Edmund Chou', with a stylized, flowing script.

Edmund Chou
Editor-in-Chief

A Necessary and Sufficient Condition for an n -lattice To Be Hamiltonian

Alex Tang

Formally, a graph $G(V, E, \psi)$ is defined to be an ordered triple that consists of a set of vertices $V(G)$, a set of edges $E(G)$, and an incidence function ψ that maps each edge to an ordered pair of vertices. Intuitively, one can think of a graph as a diagram of points, representing vertices, and lines or curves, representing edges. For example, Fig. 1 is a diagram of a graph. If two vertices are next to each other, like a and b , we say that they are adjacent. We also say that the edge connecting those two vertices are incident to them. In Fig. 1, F is incident to a and b . Additionally, a subgraph H of a graph G is graph where $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$. The statement, H is a subgraph of G , is denoted $H \subseteq G$. A spanning subgraph $S \subseteq G$ is a subgraph where $V(S) = V(G)$. Although, there is much more to be said about graphs in general, we will not discuss them here.

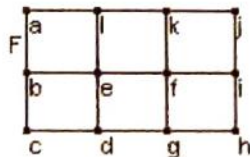


Figure 1

Here we will focus on a specific type of graph which I term an n -lattice. One can intuitively think of an n -lattice as a rectangularly shaped graph. For example, the graph represented in Fig. 1 is a 2-lattice. Similarly, a diagram shaped like a rectangular prism can represent a 3-lattice, and analogous representations can be constructed for n -lattices where $n \geq 4$. Although, this intuitive representation of n -lattices maybe useful, to work with them mathematically, we need to formally define an n -lattice.

Definition. n -lattice

For $n \geq 2$, an n -lattice L^n is a graph in which each vertex, denoted by $v[t_1, \dots, t_n]$, is given by an ordered n -tuple $(t_1, \dots, t_n)$ where $t_1, \dots, t_n \in \mathbb{N}$, any two vertices $v[t_{a1}, \dots, t_{an}] \in V(L^n)$ and $v[t_{b1}, \dots, t_{bn}] \in V(L^n)$ are adjacent if and only if $t_{az} = t_{bz}$ for all $z \in \{1, \dots, n\}$ except for one t_{ax} and one t_{bx} such that $t_{ax} = t_{bx} \pm 1$, and $|V(L^n)| = M_1 \cdots M_n$ where $M_z = \max(t_z)$ for all $z \in \{1, \dots, n\}$. For $n < 2$, n -lattices are not defined.

Using this definition, we can prove an interesting theorem about n -lattices: that they are hamiltonian if and only if they contain an even number of vertices. When we say a graph is hamiltonian, we mean it contains a hamiltonian cycle. A cycle is a sequence of vertices and edges, $v_1 e_1 v_2 \dots v_{k-1} e_{k-1} v_k$ (sometimes denoted $v_1 \dots v_k$ or $e_1 \dots e_{k-1}$ for short), where e_i is incident to v_i and v_{i+1} for any $i \in \{1, \dots, k\}$ and each vertex is distinct except for v_1 and v_k , which are the same. A hamiltonian cycle is a cycle that contains all the vertices of a graph. With these preliminaries completed, we can state and begin to prove the theorem mentioned above.

Theorem 1. Any L^n is hamiltonian if and only if $|V(L^n)|$ is even. ($|V(L^n)|$ denotes the number vertices in L^n)

To prove that $|V(L^n)|$ is even if L^n is hamiltonian, we need to use the concept of bipartness.

Definition. bipartite

A graph G is bipartite if and only if there exist two bipartitions $X \subseteq V(G)$ and $Y \subseteq V(G)$ such that no vertices in X are adjacent, no vertices in Y are adjacent, and $X \cup Y = V(G)$.

In addition, we also need the following lemma.

Lemma 1. *If a graph G is bipartite, then $|V(C)|$ is even for any cycle $C \subseteq G$.*

Proof: Let X and Y be the bipartitions of G and $C = v_0 v_1 \dots v_k v_0$ be any cycle in C . Without loss of generality, assume $v_0 \in X$. Since v_0 and v_1 are adjacent and $X \cup Y = V(G)$, then v_1 must be a element of Y . Likewise, $v_2 \in X$, $v_3 \in Y$, and so on. In general, z is even if $v_z \in X$, and z is odd if $v_z \in Y$. Because v_k is adjacent to v_0 , it must be true that $v_k \in Y$ and k is even. Since $|V(C)| = k$, it can be concluded that $|V(C)|$ is even. QED.

By lemma 1, if L^n is bipartite, any cycle it contains, including hamiltonian cycles, must have an even number of vertices. Since the number of vertices in a hamiltonian cycle of L^n is equal to the number of vertices in L^n , if we can show that L^n is bipartite, it follows that $|V(L^n)|$ is even. This is done below.

Theorem 1.

Proof: $\Rightarrow$ Let $X = \{v[t_1, \dots, t_n] \in V(L^n) \mid (t_1 + \dots + t_n) \bmod 2 = 0\}$, and let $Y = \{v[t_1, \dots, t_n] \in V(L^n) \mid (t_1 + \dots + t_n) \bmod 2 = 1\}$. Clearly, any $v[t_1, \dots, t_n] \in V(L^n)$ is a vertex of either X or Y , so the condition $X \cup Y = V(L^n)$ is satisfied. Now, by contradiction, it can be proven that no vertices in X are adjacent. Suppose two vertices $v[t_{a1}, \dots, t_{an}] \in X$ and $v[t_{b1}, \dots, t_{bn}] \in X$ are adjacent. Since $v[t_{a1}, \dots, t_{an}] \in X$, it must be true that $(t_{a1} + \dots + t_{an}) \bmod 2 = 0$. Because $v[t_{a1}, \dots, t_{an}]$ and $v[t_{b1}, \dots, t_{bn}]$ are adjacent, it must be true that $(t_{b1} + \dots + t_{bn}) = (t_{a1} + \dots + t_{an} \pm 1) \bmod 2 = 1$. However, since $v[t_{b1}, \dots, t_{bn}] \in X$, it can be concluded that $(t_{b1} + \dots + t_{bn}) \bmod 2 = 0$. Since $(t_{b1} + \dots + t_{bn}) \bmod 2 = 1$ and $(t_{b1} + \dots + t_{bn}) \bmod 2 = 0$ cannot both be true, the supposition must be false. Therefore, no two vertices in X are adjacent. By a similar argument, it can be shown that no two vertices in Y are adjacent. Thus, X and Y constitute two bipartitions of L^n , implying that L^n is bipartite. By lemma 2, $|V(C)|$ is even for any hamiltonian cycle $C \subseteq L^n$. Therefore, $|V(L^n)| = |V(C)|$ is even. QED.

By induction, we will now prove the converse of the above: L^n is hamiltonian if $|V(L^n)|$ is even. First, we must prove it for the case that $n=2$. This is done in the following lemma.

Lemma 2. L^2 is hamiltonian if $|V(L^2)|$ is even.

Proof: This lemma shall be proven graphically in an intuitive manner. Obviously, Fig. 1 represents a 2-lattice with an even number of vertices. A hamiltonian cycle of the graph

in Fig. 1 is shown in Fig. 2. One can find a similar hamiltonian cycle in any 2-lattice with an even number of vertices by tracing a path similar to the one shown in Fig. 2. This is true because one always travels up even columns of vertices and down odd ones. Since the diagram of any 2-lattice with an even number of vertices must have at least one side with an even number of vertices, one can always trace out such a hamiltonian cycle by placing that side on the bottom of the diagram. Consequently, any L^2 contains a hamiltonian cycle and is therefore hamiltonian, if $|V(L^2)|$ is even. QED.

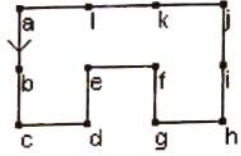


Figure 2

To complete the proof of theorem 1, we need to use the notion of isomorphism. If two graphs are isomorphic, one can think of them as being essentially the same graph. A formal definition of isomorphism is given below.

Definition. isomorphic

Two graphs G and H are isomorphic if and only if there exist two bijections $f: V(G) \rightarrow V(H)$ and $g: E(G) \rightarrow E(H)$ such that, for any two vertices $u \in V(G)$ and $v \in V(G)$ and any edge $e \in E(G)$, e is incident to u and v if and only if $g(e)$ is incident to $f(u)$ and $f(v)$. The ordered pair $\phi = (f, g)$ is called an isomorphism from G to H .

To finish the proof, we also need another lemma which involves the subgraph of $S \subseteq L^n$ created by deleting from L^n any edge incident to both a vertex $v[t_1, \dots, t_n]$ and the vertex adjacent to it of the form $v[t_1, \dots, t_{n-1}]$, except where $(t_1 = 1 \text{ and } t_n \text{ is odd})$ or $(t_1 = M_{n-1} \text{ and } t_n \text{ is even})$. This lemma also involves a function $f: V(S) \rightarrow V(L^{n-1})$ given by $f(v[s_1, \dots, s_n]) = u[t_1, \dots, t_{n-1}]$ where $t_{n-1} = (s_{n-1} + M_{n-1}(s_n - 1))(s_n \bmod 2) + (2M_{n-1} + 1 - s_{n-1} + M_{n-1}(s_n - 2))(1 - s_n \bmod 2)$ and $s_z = t_z$ for all $z \in \{1, \dots, n-2\}$. With the subgraph S and the function f defined, we can now state the following lemma.

Lemma 3. S is isomorphic to some L^{n-1} .

Proof:

It can be shown that any two vertices $v[s_{a1}, \dots, s_{an}] \in S$ and $v[s_{b1}, \dots, s_{bn}] \in S$ are adjacent if and only if their corresponding vertices $f(v[s_{a1}, \dots, s_{an}])$ and $f(v[s_{b1}, \dots, s_{bn}])$ are adjacent. However, because the proof of this statement involves much unsightly algebra, it shall be left as an exercise to the reader. (hint: Separate the problem into all the cases in which $v[s_{a1}, \dots, s_{an}]$ and $v[s_{b1}, \dots, s_{bn}]$ are adjacent and all the cases in which they are not.) Now, given that the above statement is true, we can define a function $g: E(S) \rightarrow E(L^{n-1})$ that maps the edge incident to each two vertices $v[s_{a1}, \dots, s_{an}]$ and $v[s_{b1}, \dots, s_{bn}]$ to the vertices $f(v[s_{a1}, \dots, s_{an}])$ and $f(v[s_{b1}, \dots, s_{bn}])$. Obviously, any edge $e \in E(S)$ is incident to $v[s_{a1}, \dots, s_{an}]$ and $v[s_{b1}, \dots, s_{bn}]$ if and only if $g(e) \in E(L^{n-1})$ is incident to $f(v[s_{a1}, \dots, s_{an}])$ and $f(v[s_{b1}, \dots, s_{bn}])$. Therefore, (f, g) is an isomorphism from S to L^{n-1} . Hence, S and L^{n-1} must be isomorphic. QED.

Now, we can complete the proof of theorem 1.

Theorem 1.

Proof: $\Leftarrow$ By lemma 2, the theorem is true for the case where $n=2$. Now, on the basis of the assumption that it is true for any L^{n-1} , it can be shown to be true for any L^n . By lemma 3, the spanning subgraph $S \subseteq L^n$ is isomorphic to some L^{n-1} . Because we consider two graphs that are isomorphic to be the same, it is obvious that S is hamiltonian if L^{n-1} is hamiltonian. Since L^{n-1} is hamiltonian if $|V(L^{n-1})|$ is even, S is hamiltonian if $|V(L^{n-1})|$ is even. Obviously, L^n is hamiltonian if the spanning subgraph $S \subseteq L^n$ is hamiltonian. Therefore, L^n is hamiltonian if $|V(L^n)| = |V(S)| = |V(L^{n-1})|$ is even. Thus, by the principle of mathematical induction, the theorem is proven. QED.

Although it has been shown here that a necessary and sufficient condition for an n -lattice L^n to be hamiltonian is that $|V(L^n)|$ is even, much is still unknown other types of lattice graphs. For further research, one might explore if there are similar conditions for lattices that are not rectangularly shaped, such as triangularly shaped ones. Additionally, only lattices with a finite number of vertices have been discussed here. One can also analyze lattices with an infinite number of vertices. In sum, there still remain many other types of lattices to investigate.

Periodic Properties of the Fibonacci Sequence Modulo m

James Chen

The Fibonacci sequence has intrigued mathematicians since it was introduced over 800 years ago by Leonardo of Pisa. In his book, *Liber Abaci*, which he wrote under the pseudonym Fibonacci, he introduced the following problem:

A man bought a pair of rabbits. How many pairs of rabbits will there be after each month if it is assumed that every month each pair produces one new pair, which begins to bear young two months after its own birth?

The answer to this famous problem is the following sequence of numbers (beginning with the first month): 1, 1, 2, 3, 5, ... Each term in this sequence is the sum of the two previous terms and is known as a recurrent sequence. Today, the Fibonacci sequence is usually defined as follows:

$$F_0 = 0; F_1 = 1; F_n = F_{n-1} + F_{n-2}$$

The numbers in this sequence become large very quickly, (the hundredth Fibonacci number has 21 digits!) but by reducing the sequence modulo m , we are able to keep the terms of the sequence less than or equal to $(m-1)$.

The table below shows the resulting sequences for various m when the Fibonacci sequence is taken mod m . From the table, we see that the resulting sequences are periodic. In this article, we will establish a relationship between m and the period of the sequence. Specifically, we will show that for composite m , the period is the lowest common multiple of the periods of the prime factors of m . Another interesting property that we will investigate is that, if F_0 and F_1 are changed, the period of the new sequence divides the period of the basic Fibonacci sequence.

m	period	Fibonacci sequence mod m
2	3	0,1,1,0,1,...
3	8	0,1,1,2,0,2,2,1,0,1,1,2...
4	6	0,1,1,2,3,1,0,1,1,2...
5	20	0,1,1,2,3,0,3,3,1,4,0,4,4,3,2,0,2,2,4,1,0,1,1,2...
6	24	0,1,1,2,3,5,2,1,3,4,1,5,0,5,5,4,3,1,4,5,3,2,5,1,0,1,1,2...
7	16	0,1,1,2,3,5,1,6,0,6,6,5,4,2,6,1,0,1,1,2...
8	12	0,1,1,2,3,5,0,5,5,2,7,1,0,1,1,2...
9	24	0,1,1,2,3,5,8,4,3,7,1,8,0,8,8,7,6,4,1,5,6,2,8,1,0,1,1,2...
10	60	0,1,1,2,3,5,8,3,1,4,5,9,4,3,7,0,7,7,4,1,5,6,1,7,8,5,3,8,1,9,0,9,9,8,7,5,2,7,9,6,5,1,6,7,3,0,3,3,6,9,5,4,9,3,2,5,7,2,9,1,0,1,1,2...

Definitions and Conventions Used

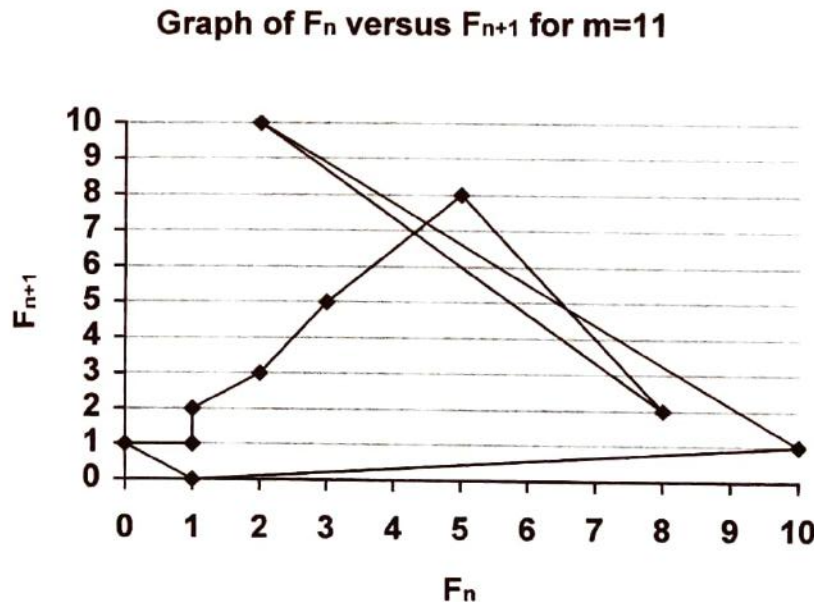
1. F_n denotes the n^{th} term of the Fibonacci sequence with $F_0=0$; $F_1=1$; and $F_n = F_{n-1} + F_{n-2}$
2. The period modulo m of the Fibonacci sequence (with $F_0=0$ and $F_1=1$), denoted by $h(m)$ or h , is the smallest positive integer for which $F_{n+h} \equiv F_n$ is satisfied for all $n = 1, 2, 3, \dots$
3. R_n denotes the n^{th} term of any recurrent series where $R_0 = a$, $R_1 = b$, and $R_n = R_{n-1} + R_{n-2}$.
4. The period modulo m of any recurrent sequence (with $R_0=a$ and $R_1=b$), denoted by $k(m)$ or k , is the smallest positive integer for which $R_{n+k} \equiv R_n$ is satisfied for all $n = 1, 2, 3, \dots$
5. In this article, the word period is used to mean both the sequence of numbers repeated and the number of terms in the repeated sequence.

In this paper we will prove the following four conjectures:

1. When reduced modulo m , the Fibonacci sequence is always periodic. That is, for all m , h exists.
2. The period of composite m is the lowest common multiple of the periods of the prime factors of m .
3. If zeros occur in the sequence before a period is repeated, then its subscript is a factor of the period.
4. The period of any recurrent sequence reduced modulo m must divide the period of the Fibonacci sequence reduced modulo m .

Theorem 1. $\{F_n \pmod{m}\}$ forms a simply periodic sequence. That is, the sequence is periodic and repeats by returning to its starting values.

Proof: Let us first consider a specific case with $m=11$. The sequence generated is thus: 0,1,1,2,3,5,8,2,10,1,0,1,1,..., which has a period of 10. If we plot F_n versus F_{n+1} , we get the following graph:



By the definition of the Fibonacci numbers, for each point on the graph, there is only one point that can follow. Thus, if we revisit any point on this graph, the sequence becomes periodic. Since there are only $m \times m$ lattice points on the graph, some point must be revisited within (m^2+1) steps. In fact, cycle lengths have an upper bound of m^2 .

From the definition of Fibonacci numbers, we have $F_{n-1} = F_{n+1} - F_n$. So if $F_{n+1} \equiv F_{s+1} \pmod{m}$ and $F_n \equiv F_s \pmod{m}$ then $F_{n-1} \equiv F_{s-1} \pmod{m}$, $\dots$, $F_{t-s+1} \equiv F_1 \pmod{m}$, and $F_{t-s} \equiv F_0 \pmod{m}$. Thus, each period must begin with F_0 . Therefore, we can conclude that $\{F_n \pmod{m}\}$ forms a simply periodic sequence. QED.

Theorem 2. If the prime factorization of m is $m = \prod p_i^{e_i}$ and if h_i denotes the length of the period of $\{F_n \pmod{p_i^{e_i}}\}$, then $h = \text{lcm}[h_i]$, the least common multiple of all the h_i .

Proof: Since h_i is the length of the period of $\{F_n \pmod{p_i^{e_i}}\}$, this means that the sequence $\{F_n \pmod{p_i^{e_i}}\}$ repeats only after blocks of length ch_i where c is some integer. Thus the term F_n only repeats after ch_i terms. A well-known property of moduli is if $a \equiv b \pmod{mk}$ then $a \equiv b \pmod{m}$ where k is any positive integer. Using this property, we know that h is the period of $\{F_n \pmod{m}\}$ thus implies that $F_n \pmod{p_i^{e_i}}$ repeats after h terms for all values of i . So, for all i , h is of the form ch_i . Therefore, $h = \text{lcm}[h_i]$. QED.

Theorem 3. If n is the smallest positive integer for which $F_n \equiv 0 \pmod{m}$, then $F_{kn} \equiv 0 \pmod{m}$ for $k=1, 2, 3, \dots$

Proof: In order to complete this proof we must utilize two well known relationships of Fibonacci numbers that are simple to prove:

(i) $\text{gcf}(F_n, F_{n+1}) = 1$

(ii) $F_{n+t} = F_{n+1}F_t + F_nF_{t-1}$

To prove the (i), let $\text{gcf}(F_n, F_{n+1}) = d$. By the definition of gcf , $d \mid F_n$ and $d \mid F_{n+1}$. Thus, $d \mid (F_{n+1} - F_n)$ or $d \mid (F_{n-1})$. If we repeat this process, we will eventually reach the conclusion that $d \mid F_1$. But $F_1 = 1$, so $d = 1$. Thus $\text{gcf}(F_n, F_{n+1}) = 1$.

To prove (ii), we do a proof by strong induction on t .

Base Case: ($t=1$)

$$\begin{aligned} F_{n+1} &= F_{n+1}F_1 + F_nF_0 \\ &= F_{n+1}(1) + F_n(0) \\ &= F_{n+1} \end{aligned}$$

Inductive Step:

Since we will use strong induction to prove the theorem, we assume that the statement is true for $t=1, 2, 3, \dots, k$. We want to show that it is true for $(k+1)$. More specifically, we will assume it to be true for k and $(k-1)$:

$$\begin{aligned} F_{n+k} &= F_{n+1}F_k + F_nF_{k-1} \\ F_{n+k-1} &= F_{n+1}F_{k-1} + F_nF_{k-2} \end{aligned}$$

We wish to show:

$$\begin{aligned} F_{n+k+1} &= F_{n+1}F_{k+1} + F_nF_k \\ F_{n+k+1} &= F_{n+k} + F_{n+k-1} \\ &= F_{n+1}F_k + F_nF_{k-1} + F_{n+1}F_{k-1} + F_nF_{k-2} \\ &= F_{n+1}(F_k + F_{k-1}) + F_n(F_{k-1} + F_{k-2}) \\ &= F_{n+1}F_{k+1} + F_nF_k \end{aligned}$$

Thus, by strong induction, we have shown (ii) to be true.

Now that we have established (i) and (ii), we return to our proof of theorem 3. If $F_n \equiv 0 \pmod{m}$ and $F_j \equiv 0 \pmod{m}$, then we can show that $F_{i+j} \equiv 0 \pmod{m}$ as follows:

$$F_{n+t} \equiv F_{n+1}F_t + F_nF_{t-1} \pmod{m} \quad [\text{follows directly from (ii)}]$$

$$\begin{aligned}F_{i+j} &= F_{i+1}F_j + F_iF_{j+1} \pmod{m} \\F_{i+j} &= F_{i+1}(0) + (0)F_{j+1} \pmod{m} \\F_{i+j} &= 0 \pmod{m}\end{aligned}$$

Next, (assuming $i \leq j$), we show that $F_{i-j} \equiv 0 \pmod{m}$

$$F_{n+i} \equiv F_{n+1}F_i + F_nF_{i+1} \pmod{m} \quad [\text{follows directly from (ii)}]$$

If we let $i = (n+1)$, we get:

$$F_i \equiv F_{n+1}F_i + F_nF_{i+1} \pmod{m}$$

Since $F_i \equiv 0 \pmod{m}$ and $F_n \equiv 0 \pmod{m}$, $F_{n+1}F_i \equiv 0 \pmod{m}$. Also $\gcd(F_{n+1}, F_n) = 1$, so $F_i \equiv 0 \pmod{m}$. But $F_i = F_{i-j}$, so $F_{i-j} \equiv 0 \pmod{m}$.

From this, we can conclude that all x for which $F_x \equiv 0$ must be a multiple of n . That is, $F_{kn} \equiv 0 \pmod{m}$ for $k=1,2,3,\dots$. From the data tables, we observe that while $n=h$ in some cases ($m=4,11,19,\dots$), $n < h$ for many other cases ($m=3,5,6,7,8,9,10,\dots$). QED.

Theorem 4. *The period mod m of any recurrent sequence must be a factor of the period mod m of the Fibonacci sequence. That is, for all m , $h(m) \mid k(m)$.*

Proof. If the first two terms of a recurrent series are a and b , we note that the sequence becomes: $a, b, a+b, 2a+b, 3a+2b, 5a+3b, \dots$. The pattern that develops is the following:

$$R_n = aF_{n-1} + bF_n$$

The above equation can be easily verified using induction.

Next, we prove that $R_{n+h} \equiv R_n \pmod{m}$

$$\begin{aligned}R_{n+h} &\equiv aF_{n+h-1} + bF_{n+h} \pmod{m} \\&\equiv aF_{n-1} + bF_n \pmod{m} \\&\equiv R_n \pmod{m}\end{aligned}$$

We note that since F_0 and F_1 were not involved in the proof of Theorem 1, the theorem is actually valid for all recurrent sequences. Therefore, all recurrent sequences must be periodic mod m for all m . If there exists $k < h$, then $k \mid h$ because $R_{n+h} \equiv R_n \pmod{m}$. QED.

This article has touched upon the basic periodic properties of the Fibonacci sequence mod m . We know that the Fibonacci sequence mod m must eventually become periodic for all m and that the period of composite m is related to the periods of the prime factors of m . We also looked briefly at other recurrent sequences that are constructed in a manner similar to the Fibonacci sequence. Hopefully this brief article has whetted the reader's appetite to further examine these sequences. Below are a few extensions that the reader may wish to explore:

1. In Theorem 1, we proved that the every period must begin with a 0. In Theorem 3, we showed that the zeros in the sequence occur at equal intervals. However, the first 0 does not always indicate the beginning of a new period. For example, taking the Fibonacci sequence mod 8, we get the following sequence: 0,1,1,2,3,5,0,5,5,2,2,7,1,0,1,1,... Although $F_6 \equiv 0 \pmod{8}$, $h(8)=12$. Is there any relationship between m and the number of zeros that must be encountered before the sequence repeats?
2. Theorem 2 provides us with a way to find h for composite m . Is there a formula for finding h for prime m ?
3. In designing a (pseudo)random number generator, it is helpful to research the distribution of the numbers in the sequence. Do all the numbers $0,1,2,3,\dots,(m-1)$ have to appear? Does each number

appear an equal number of times? If not, how can we relate its probability to other characteristics of the sequence (such as s , m , etc.)?

References

- [1] Hardy, G. H. and Wright, E. M.. An Introduction to the Theory of Numbers. Oxford University Press: Oxford, 1979
- [2] Jarden, Dov. "On the Periodicity of the Last Digits of the Fibonacci Numbers." The Fibonacci Quarterly. Vol. 1, No. 4. December 1963. pp. 21-22.
- [3] Knott, Ron. "Fibonacci Numbers and the Golden Section".
<http://www.ee.surrey.ac.uk/Personal/R.Knott/Fibonacci/fib.html>
- [4] Vorob'ev, N. N. Fibonacci Numbers. Pergamon Press, Ltd. 1961
- [5] Wall, D. D. "Fibonacci Series modulo m ". American Math Monthly. Vol. 67. June-July 1960. pp. 525-532.

Minkowski's Theorem

Chia-Hao La

Mathematics is a field that consists of many areas of study, all very different from each other. Although it often seems that these areas have almost nothing in common, everything in math is in fact closely connected in a beautiful unifying structure. One example is a well-known theorem of Minkowski that, at first glance, seems to deal only with geometry. However, the theorem is intimately tied to the field of number theory.

Basic Definitions:

- 1) A region K in $\mathbb{R}^2$ is said to be convex if for any points $x_1, x_2 \in K$, the line segment joining x_1 and x_2 is contained in K .
- 2) A region K in $\mathbb{R}^2$ is symmetric about O (the origin) if and only if $\forall x \in K, -x$ is also in K . For example, in Figure 1, region A is convex but not symmetric about O . Region B is symmetric about O but not convex.

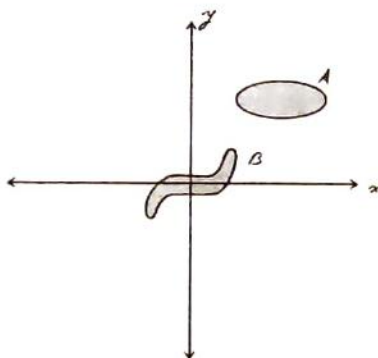


Figure 1

Now, consider the standard lattice $\mathbb{Z}^2$ and a region K which is both convex and symmetric about O . How big can K be such that it doesn't contain any lattice points other than O ? The Theorem of Minkowski states that K can have any area less than 4 and still fulfill this condition. Figure 2 is an example of such a region. If the area of K exceeds 4, then K must contain at least one non-zero lattice point.

Let us examine one special example of Minkowski's Theorem. In Figure 3, the line $y = \sqrt{2}x$ is plotted on the coordinate plane. Since the line has an irrational slope, it obviously does not pass through any lattice points besides the origin. Now let us consider a segment of this line that is

symmetric about the origin and is 200 units long. If we slowly expand the segment into a thin convex region, Minkowski's Theorem predicts that this long "hotdog" will not contain any lattice points besides the origin as long as its area is less than 4. If we continue expanding the region until its area is greater than 4, then it will contain at least one nonzero lattice point.

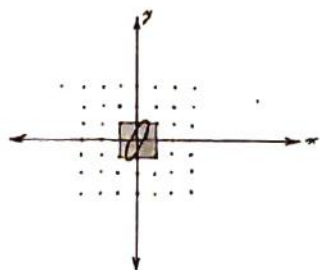


Figure 2

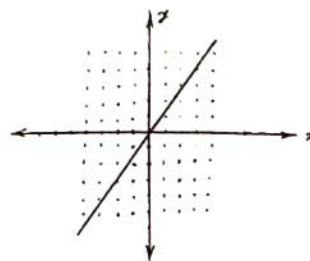


Figure 3

We are accustomed to working in the standard lattice, which is determined by the vectors $(0,1)$ and $(1,0)$. However, Minkowski's Theorem applies to any lattice in $\mathbf{R}^2$. The base vectors v_1 and v_2 of the lattice need not be perpendicular or of the same length (see Figure 4). The general form of the theorem is as follows:

Minkowski's Theorem. Let L be a lattice in $\mathbf{R}^2$ and let Δ be the area of a fundamental parallelogram defined by the vectors v_1 and v_2 . Let K be a convex region in $\mathbf{R}^2$ which is symmetric about O . If $\text{Area}(K) > 4\Delta$, then K contains a nonzero lattice point in L .

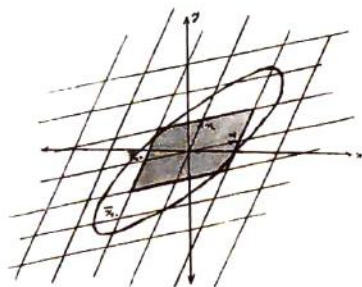


Figure 4

Proof: Consider $2L = \{2v \mid v \in L\}$

The following is an intuitive statement and is presented here without proof:

Since $\text{Area}(K) > 4\Delta$, there exist points $x_1, x_2 \in K$ such that $\overline{x_1} \neq \overline{x_2}$ and $\overline{x_1} \equiv \overline{x_2} \pmod{2L}$.

Here $\overline{x_n} = -x_n$, or the reflection of x_n about the origin. The notation in the last part of the statement means that $\overline{x_1}$ and $\overline{x_2}$ are both at the same position within their respective parallelograms in $2L$ (see Figure 4).

$x_1 \in K \rightarrow \overline{x_1} \in K$ (since K is symmetric about O)

$\overline{x_1}, x_2 \in K \rightarrow \frac{\overline{x_1} + x_2}{2} \in K \rightarrow \frac{\overline{x_1} + (-\overline{x_2})}{2} \in K$ (since K is convex)

Let, $\overline{l} = \frac{\overline{x_1} - \overline{x_2}}{2} \in K$.

$\overline{x_1} \equiv \overline{x_2} \pmod{2L} \rightarrow \overline{x_1} - \overline{x_2} \in 2L \rightarrow \frac{\overline{x_1} - \overline{x_2}}{2} = \overline{l} \in L$

$\overline{x_1} \neq \overline{x_2} \rightarrow \overline{x_1} - \overline{x_2} \neq 0 \rightarrow \overline{l} \neq 0$

$\therefore \overline{l}$ is a nonzero lattice point in L that is also in K . QED.

As a final note, not only does Minkowski's Theorem rely on certain elements of number theory for its proof, but it also has applications within number theory. For example, it can be used to show that if $p \equiv 1 \pmod{4}$ is a positive prime, then it is a sum of two integer squares. This statement is usually proven using Gaussian integers, but Minkowski's Theorem offers a simpler, more elegant proof.

References

- [1] Hardy, G.H. and Wright, E.M. An Introduction to the Theory of Numbers, Fifth Edition. Oxford University Press: New York, 1995

Special thanks to Professor Glenn Stevens of Boston University's Program in Mathematics for Young Scientists (PROMYS)

Genetics with Set Theory

Edmund Chou

The biological sciences have steadily grown to include more mathematical analysis, such as the use of set theory in genetics. Before exploring the mathematical aspects of this subject, it would be useful to briefly review some essential definitions and terminology from genetics.

A *gene* is an entity that determines hereditary, which is found in each cell of a biological organism. These basic entities are assumed to be located in certain parts of the chromosomes. Of two or more alternative forms or *alleles* of a gene, one exists at each *locus*. For simplicity, the discussion will be confined to two possible alleles for a locus.

The focus of the discussion will concentrate on a single locus and the character of the organism affected by alleles of the gene at this locus. Suppose that the gene at this locus has n alleles which will be represented by the elements of the set $\{a_1, a_2, \dots, a_n\}$. With the assumption that the locus contains only two alleles, this fact is indicated by an ordered pair $\{a_i, a_j\}$ chosen from the set of possible alleles, representing a *genotype* of $\{a_i, a_j\}$. The organism is *homozygous* if both genes are represented by the same allele $\{a_i, a_i\}$. If the alleles are different, the organism is called *heterozygous*.

Observable characteristics of an organism do not always indicate the particular genes that are present. Thus it is convenient to distinguish between the *appearance* or the *phenotype* of an organism, and the *genetic composition* or *genotype*. The heterozygous genotype $\{a_i, a_j\}$ may exhibit a trait corresponding to either the a_i or the a_j allele. The gene of the heterozygous genotype which is seen in the phenotype is called the *dominant* allele, while the allele which does not appear is called the *recessive* allele.

It is often necessary to determine the genetic constitution of a group in terms of a percentage or proportion of individuals belonging to each genotype. These ratios are called *genotypic frequencies*. For example, if the genotype $\{a_1, a_1\}$ occurs in one fifth of the individuals in a population, its genotypic frequency is 20 percent. Of course, the total of all genotypic frequencies must add up to 100 percent. A *gene frequency* is similarly defined; if a particular gene a_3 occurs in a quarter of the population, then the gene frequency for a_3 is 25 percent.

A population is said to be *mating at random* with respect to certain genetic characters when an individual with a particular genotype mates purely by chance with a member of the population who has or does not have the genotype in question. Hardy, a English mathematician, and Weinberg, a German physician, developed the Hardy-Weinberg Law in 1908 after observing a phenomenon. The law states that *in a large population after one generation of random mating the genotypic frequencies as well as the gene frequencies of a given locus remain the same for all succeeding generations provided there is no selection, immigration, or mutation* [1].

Definition. Let $G = \{\{a_{i_1}, a_{j_1}\}, \{a_{i_2}, a_{j_2}\}, \dots, \{a_{i_m}, a_{j_m}\}\}$ be any set of m genotypes at a given locus of the individuals of the parent generation. These m pairs are not necessarily distinct and involve elements selected from the set $\{a_1, a_2, \dots, a_n\}$ of alleles.

Using the definition, a mathematical operation involving set theory can be used to obtain a set of all possible genotypes arising from mating within the original set and values for stable frequencies. The first step is to form the set G' whose elements are the $2m$ genes a_k , including all duplicates, that occur in

the genotypes of set G . Next the construction of the product set $G' \times G'$ will represent the set of all ordered pairs $\{a_{i_h}, a_{j_k}\}$ formed from the elements of G' .

The product set $G' \times G'$ contains the genotypes of the first filial generation, F_1 , from the parent set of genotypes, G . The frequency of each type should be the corresponding stable frequency mentioned in the Hardy-Weinberg Law.

To illustrate, it will be assumed that only two alleles are involved in the genotypes of G , and that the set G' consists of r alleles of type a_1 and s alleles of type a_2 , and thus $r + s = m$. In order to keep track of the ordered pairs in the set $G' \times G'$, the following table is constructed:

	$r: a_1$	$s: a_2$
$r: a_1$	$r^2: a_1a_1$	$rs: a_1a_2$
$s: a_2$	$rs: a_2a_1$	$s^2: a_2a_2$

where the notation $r^2: a_1a_1$ denotes that the genotype $\{a_1, a_1\}$ occurs r^2 times in the set $G' \times G'$ and so forth. Since the total number of elements in $G' \times G'$ is $m^2 = (r + s)^2$, the following genotypic frequencies in F_1 are r^2/m^2 for type $\{a_1, a_1\}$, $2rs/m^2$ for type $\{a_1, a_2\} = \{a_2, a_1\}$, and s^2/m^2 for type $\{a_2, a_2\}$.

This process can be repeated to obtain the genotypic frequencies for the F_2 generation. The set G'' contains the $2m^2$ alleles which occur in the genotypes of $G' \times G'$. There are $2r^2 + 2rs = 2r(r + s) = 2rm$ of the a_1 type and $2s^2 + 2rs = 2s(s + r) = 2sm$ of the a_2 type. The set $G'' \times G''$ can then be constructed with the following table:

	$2rm: a_1$	$2sm: a_2$
$2rm: a_1$	$4r^2m^2: a_1a_1$	$4rsm^2: a_1a_2$
$2sm: a_2$	$4rsm^2: a_1a_2$	$4s^2m^2: a_2a_2$

which yields the genotypic frequencies $4r^2m^2/4m^4 = r^2/m^2$ for type $\{a_1, a_1\}$, $8rsm^2/4m^4 = 2rs/m^2$ for type $\{a_1, a_2\}$, and $4s^2m^2/4m^4 = s^2/m^2$ for type $\{a_2, a_2\}$ in F_2 . Therefore, it is evident that the stable genotypic frequencies explained by Hardy and Weinberg hold true.

To expand on this idea, another problem can be explored. The next illustration defines the parent generation with genotypes $G = \{\{a_1, a_2\}, \{a_3, a_4\}\}$. The genes in the set $G' = \{a_1, a_2, a_3, a_4\}$ are assumed to combine randomly and the genotypic frequencies of the F_1 and F_2 generations are to be determined.

The product set $G' \times G'$ in this problem is shown in the following array:

	a_1	a_2	a_3	a_4
a_1	a_1a_1	a_1a_2	a_1a_3	a_1a_4
a_2	a_2a_1	a_2a_2	a_2a_3	a_2a_4
a_3	a_3a_1	a_3a_2	a_3a_3	a_3a_4
a_4	a_4a_1	a_4a_2	a_4a_3	a_4a_4

There are 16 genotypes in $G' \times G'$, of which there are one each of the homozygous type $\{a_i, a_i\}$ and two each of the heterozygous type $\{a_i, a_j\}$ where $i \neq j$, $i = 1, 2, 3, 4$, and $j = 1, 2, 3, 4$. The genotypic frequencies for the F_1 generation are then $1/16$ for each type $\{a_i, a_i\}$ and $1/8$ for each type $\{a_i, a_j\}$, $i \neq j$.

This process is repeated by first forming the set G'' which is each composed of four genes a_i where $i = 1, 2, 3, 4$. The composition of the product set $G'' \times G''$ is determined by an array of 16 rows and 16 columns. There will be a total of 256 genotypes of which there are 16 each of the homozygous

type and 32 each of the heterozygous type. Thus the genotypic frequencies are $16/256 = 1/16$ for each type $\{a_i, a_i\}$ and $32/256 = 1/8$ for each type $\{a_i, a_j\}$, $i \neq j$, as predicted by the Hardy-Weinberg Law.

Concentrating on two genes a_1 and a_2 whose frequencies are denoted by f_1 and f_2 respectively, the results can be expressed generally. The quantities f_1 and f_2 can be thought of as the number of a_1 and a_2 genes in the population. Denote the corresponding genotypic frequencies by f_{11} , f_{12} and f_{22} . To summarize, the genotypes $\{a_1, a_1\}$, $\{a_1, a_2\} = \{a_2, a_1\}$, and $\{a_2, a_2\}$ will respectively have the frequencies $f_{11} = f_1^2$, $f_{12} = f_1 f_2 + f_2 f_1 = 2 f_1 f_2$, and $f_{22} = f_2^2$.

These generalizations are useful in solving problems in which the Hardy-Weinberg Law applies. Suppose a flock of a certain species of fowl consists of 117 with blue feathers, 191 with black feathers, and 16 with white feathers. The black and white feathers are the phenotypes corresponding to the homozygous genotypes $\{b, b\}$ and $\{w, w\}$, and the blue feather phenotype corresponds to the heterozygous genotype. Assuming that the Hardy-Weinberg Law is valid in this case, find the genotypic and gene frequencies.

The genotypic frequencies are easily determined with the given information. Since the total number of birds is 324, $f_{bb} = 191/324 \approx 0.59$, $f_{ww} = 16/324 \approx 0.05$, and $f_{bw} = 117/324 \approx 0.36$. Two equations $f_b^2 = 0.59$ and $f_w^2 = 0.05$ are used to determine the gene frequencies, which yield $f_b \approx 0.77$ and $f_w \approx 0.22$. Note that $f_b + f_w = 0.99$, which is not too far from 1.00. The error is a result of rounding off, and a further check from the equation $2f_b f_w = 0.36$ indicates that $f_w = 0.36/2(0.77) = 0.23$.

References

- [1] Nahikian, Howard M. A Modern Algebra for Biologists. The University of Chicago Press: Chicago, 1964.
- [2] Searle, S. R. Matrix Algebra for the Biological Sciences. John Wiley & Sons, Inc.: New York, 1966.

Some Remarkable Inequalities

Regina Eum

Over the years, mathematicians have discovered many interesting inequalities. Two of the most famous ones are the Arithmetic Mean-Geometric Mean (AM-GM) inequality and the Cauchy-Schwartz Inequality. The former states that the average of any set of numbers $a_1, a_2, \dots, a_n$ is greater than or equal to the n^{th} root of their product; this inequality is a specific case of the Power Mean Inequality. The Power Mean, $P(r)$, of a set of numbers $a_1, a_2, \dots, a_n$ is $P(r) = [(1/n)(a_1^r + a_2^r + \dots + a_n^r)]^{(1/r)}$, where $a > 0$, $r \neq 0$, and $|r|$ is less than infinity. The Power Mean Inequality states that if we have two powers x and y , where $x > y$, $P(x) \geq P(y)$. (Equality results if and only if $a_1 = a_2 = \dots = a_n$.) Since the Arithmetic mean is $P(1)$ and the geometric mean is $P(0)$, the AM-GM inequality follows naturally from the Power Mean Inequality. The Cauchy-Schwartz Inequality states that for any set of numbers $a_1, a_2, \dots, a_n; b_1, b_2, \dots, b_n$, $(a_1^2 + a_2^2 + \dots + a_n^2)(b_1^2 + b_2^2 + \dots + b_n^2) \geq (a_1b_1 + a_2b_2 + \dots + a_nb_n)^2$.

These two inequalities are powerful weapons for attacking problems involving inequalities, and often lead to the creation of new theorems such as the ones below.

Theorem 1. For any positive numbers $a_1, a_2, \dots, a_n; b_1, b_2, \dots, b_n$, $[(a_1 + b_1)^2 + (a_2 + b_2)^2 + \dots + (a_n + b_n)^2]^{1/2} \leq [a_1^2 + a_2^2 + \dots + a_n^2]^{1/2} + [b_1^2 + b_2^2 + \dots + b_n^2]^{1/2}$.

Proof: Note that $(a_1 + b_1)^2 + (a_2 + b_2)^2 + \dots + (a_n + b_n)^2 = a_1(a_1 + b_1) + a_2(a_2 + b_2) + \dots + a_n(a_n + b_n) + b_1(a_1 + b_1) + b_2(a_2 + b_2) + \dots + b_n(a_n + b_n)$. By the Cauchy-Schwartz Inequality, we know that $(a_1 + b_1)a_1 + (a_2 + b_2)a_2 + \dots + (a_n + b_n)a_n \leq [(a_1 + b_1)^2 + (a_2 + b_2)^2 + \dots + (a_n + b_n)^2]^{1/2} [a_1^2 + a_2^2 + \dots + a_n^2]^{1/2}$. If we instead let $b_1 = y_1, b_2 = y_2, \dots, b_n = y_n$, the Cauchy-Schwartz Inequality yields $(a_1 + b_1)b_1 + (a_2 + b_2)b_2 + \dots + (a_n + b_n)b_n \leq [(a_1 + b_1)^2 + (a_2 + b_2)^2 + \dots + (a_n + b_n)^2]^{1/2} [b_1^2 + b_2^2 + \dots + b_n^2]^{1/2}$. Adding the two inequalities we gained using the Cauchy-Schwartz Inequality gives us $(a_1 + b_1)^2 + (a_2 + b_2)^2 + \dots + (a_n + b_n)^2 \leq [(a_1 + b_1)^2 + (a_2 + b_2)^2 + \dots + (a_n + b_n)^2]^{1/2} [(a_1^2 + a_2^2 + \dots + a_n^2) + (b_1^2 + b_2^2 + \dots + b_n^2)]^{1/2}$. If we divide both sides by $[(a_1 + b_1)^2 + (a_2 + b_2)^2 + \dots + (a_n + b_n)^2]^{1/2}$, we obtain the desired inequality: $[(a_1 + b_1)^2 + (a_2 + b_2)^2 + \dots + (a_n + b_n)^2]^{1/2} \leq [a_1^2 + a_2^2 + \dots + a_n^2]^{1/2} + [b_1^2 + b_2^2 + \dots + b_n^2]^{1/2}$. QED.

Theorem 2. Let $a_1, a_2, \dots, a_n; b_1, b_2, \dots, b_n; \dots; m_1, m_2, \dots, m_n$ be k sequences of positive numbers, and $r, s, \dots, w$ be k positive numbers such that $r + s + \dots + w = 1$. Then $a_1^r b_1^s \dots m_1^w + a_2^r b_2^s \dots m_2^w + \dots + a_n^r b_n^s \dots m_n^w \leq (a_1 + a_2 + \dots + a_n)^r (b_1 + b_2 + \dots + b_n)^s \dots (m_1 + m_2 + \dots + m_n)^w$.

Proof: Let $a_1 + a_2 + \dots + a_n = a, b_1 + b_2 + \dots + b_n = b, \dots, m_1 + m_2 + \dots + m_n = m$. If the inequality of the problem is divided by $a^r b^s \dots m^w$, an equivalent inequality is found: $[(a_1/a)^r (b_1/b)^s \dots (m_1/m)^w] + [(a_2/a)^r (b_2/b)^s \dots (m_2/m)^w] + \dots + [(a_n/a)^r (b_n/b)^s \dots (m_n/m)^w] \leq 1$. Through the AM-GM inequality, we obtain $(a_k/a)^r (b_k/b)^s \dots (m_k/m)^w \leq [(r(a_k/a) + s(b_k/b) + \dots + w(m_k/m))/(r + s + \dots + w)]^{r+s+\dots+w}$. Since $r + s + \dots + w = 1$, the right side of the inequality equals $r(a_k/a) + s(b_k/b) + \dots + w(m_k/m)$. The sum of these inequalities for $k = 1, 2, \dots, n$ yields $[(a_1/a)^r (b_1/b)^s \dots (m_1/m)^w] + [(a_2/a)^r (b_2/b)^s \dots (m_2/m)^w] + \dots + [(a_n/a)^r (b_n/b)^s \dots (m_n/m)^w] \leq [ra_1/a + sb_1/b + \dots + wm_1/m] + [ra_2/a + sb_2/b + \dots + wm_2/m] + \dots + [ra_n/a + sb_n/b + wm_n/m]$. We can rearrange the terms on the right side of the inequality to get $(r/a)(a_1 + a_2 + \dots + a_n) + (s/b)(b_1 + b_2 + \dots + b_n) + \dots + (w/m)(m_1 + m_2 + \dots + m_n)$, which

equals $r(a/a) + s(b/b) + \dots + w(m/m) = r + s + \dots + w = 1$, which is what we wished to show. Equality results only if $a_1:b_1:\dots:m_1 = a_2:b_2:\dots:m_2 = \dots = a_n:b_n:\dots:m_n$. QED.

Theorem 3. If $x_1, x_2, \dots, x_n$ are n positive numbers and g is their geometric mean, then $(1 + x_1)(1 + x_2) \dots (1 + x_n) \geq (1 + g)^n$.

Proof: We can substitute the n two-number sequences $1, x_1; 1, x_2; \dots; 1, x_n$ and $r = s = \dots = w = 1/n$ into the equation provided by Theorem 2 to obtain the solution to the generalized problem. Substitution gives us $1 + x_1^{1/n} x_2^{1/n} \dots x_n^{1/n} \leq (1 + x_1)^{1/n} (1 + x_2)^{1/n} \dots (1 + x_n)^{1/n}$. We can raise both sides to the n^{th} power and substitute g for $(x_1 x_2 \dots x_n)^{1/n}$ to get $(1 + g)^n \leq (1 + x_1)(1 + x_2) \dots (1 + x_n)$, which can be re-written as $(1 + x_1)(1 + x_2) \dots (1 + x_n) \geq (1 + g)^n$. QED.

Theorem 4. For any positive number k , $(k + 1^n)(k + 2^n) \dots (k + n^n) \geq (4k(n!))^{n/2}$ and $(k + 1^n)(k + 2^n) \dots (k + n^n) \geq (k + n!)^n$.

Proof: Solution 1. By the AM-GM inequality, we know that for any integer m less than or equal to n , $(k + m^n) \geq 2(k(m^n))^{1/2}$. Thus the inequality becomes $(k + 1^n)(k + 2^n) \dots (k + n^n) \geq 2(k(1^n))^{1/2} 2(k(2^n))^{1/2} \dots 2(k(n^n))^{1/2} = (4k(n!))^{n/2}$.
 Solution 2. $(k + 1^n)(k + 2^n) \dots (k + n^n) = (k^n)(k + 1^n)(k + 2^n) \dots (k + n^n)/(k^n) = (k^n)(1 + (1/k)^n)(1 + (2/k)^n) \dots (1 + (n/k)^n) \geq (k^n)(1 + ((1/k)^n(2/k)^n \dots (n/k)^n)^{1/n})^n$ by Theorem 3.
 Since $(k^n)(1 + ((1/k)^n(2/k)^n \dots (n/k)^n)^{1/n})^n = (k^n)(1 + ((1 \times 2 \times \dots \times n)^n/k^n)^{1/n})^n = (k^n)(1 + (n!/k))^n = (k^n)((k + n!)/k)^n \geq (k + n!)^n$. QED.
 Note: $(k + n!)^n = (4k(n!))^{n/2}$ when $k = n!$ and $(k + n!)^n > (4k(n!))^{n/2}$ when $k > n!$

References

- [1] Beckenbach, Edwin and Bellman, Richard. *An Introduction to Inequalities*. Mathematical Association of America, 1961, pp. 71.
- [2] *Crux Mathematicorum*. Vol. 19, No.3. Canadian Mathematical Society, March 1993.
- [3] *Crux Mathematicorum*. Vol. 19, No.4. Canadian Mathematical Society, April 1993.
- [4] *Crux Mathematicorum*. Vol. 20, No.1. Canadian Mathematical Society, January 1994.
- [5] *Crux Mathematicorum*. Vol. 20, No.4. Canadian Mathematical Society, April 1994.
- [6] Kazarinoff, Nicholas D. *Geometric Inequalities*. Mathematical Association of America, 1961.
- [7] *Mathematics Magazine*. Vol. 69, No. 5. Mathematics Association of America, December 1996.
- [8] *Mathematics Magazine*. Vol. 70, No. 4. Mathematics Association of America, October 1997.
- [9] Shklarsky, D.O., Chentzov, N.N. and Yaglom, I.N. The USSR Olympiad Problem Book. Dover, 1993.

Chaos Theory and Newton's Method

Nelson Uhan

Chaos theory, known for its exotic applications in the biological sciences, also can be found lurking in the depths of our calculus textbooks, particularly when we learn about Newton's method for approximating roots. Newton's method is used to find the zeros of a function when elementary algebraic methods cannot be used. For example, the zeros of the function $f(x) = x^3 - x + 1$ cannot be found by factoring.

Newton's method for approximating zeros is based on the assumption that the graph a function f and its tangent line at $(x_1, f(x_1))$ both cross the x -axis at nearly the same point. Since we can easily calculate the x -intercept for this tangent line, we use it as our second, and hopefully better estimate for the zero of f . The tangent passes through the point $(x_1, f(x_1))$ with a slope of $f'(x_1)$. In point-slope form, the equation of the tangent line is therefore $y - f(x_1) = f'(x_1)(x - x_1)$ or $y = f'(x_1)(x - x_1) + f(x_1)$.

Letting $y = 0$ and solving for x , we have $x = x_1 - \frac{f(x_1)}{f'(x_1)}$. Thus, from our initial guess we arrive at a new estimate, x_2 where $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$.

If we reiterate this equation over and over, we hopefully will be arriving at better and more accurate estimates for the zeros of function f . However, Newton's method is not fail-safe. For example, take the function $f(x) = x^{1/3}$ with an initial guess of 0.1

Since $f(x) = \frac{1}{3}x^{-2/3}$, the iterative formula is $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^{1/3}}{\frac{1}{3}x_n^{-2/3}} = x_n - 3x_n = -2x_n$

Table of Iterations:

n	x_n	$f(x_n)$	$f'(x_n)$	$\frac{f(x_n)}{f'(x_n)}$	$x_n - \frac{f(x_n)}{f'(x_n)}$
1	0.10000	0.46416	1.54720	0.30000	-0.20000
2	-0.20000	-0.58480	0.97467	-0.60000	0.40000
3	0.40000	0.73681	0.61401	1.20000	-0.80000
4	-0.80000	-0.92832	0.38680	-2.40000	1.60000

Newton's method here seems to fail to converge to a specific root. This will occur no matter what the value of x_1 is. However, another problem with Newton's method deals with selecting an appropriate initial guess. Not all choices of starting points converge quickly on a specific root.

The potential problems increase when Newton's method is applied to functions of a complex variable. Equations involving complex numbers bring in two dimensions: each complex variable, z , can be thought of as having two components; whereas real numbers can be represented as points on a line, complex numbers must be located on a plane.

A polynomial equation such as $z^4 - 1 = 0$ has as many roots as the highest power to which z is raised in the equation. With this particular equation, there are four complex roots. In terms of Newton's method, these roots act as centers of an attractive force field that draws a sequence of guesses closer and closer to one of the roots. When Newton's method is used to find a specific root, the solver hopes that the chosen starting point leads quickly to the appropriate root. However, the method fails when the

chosen point happens to fall on a boundary separating regions “controlled” by different roots. Failure may also strike when, for some starting values, the procedure gets sidetracked and ends up going nowhere, like the example shown previously.

Computer graphics provides a way to picture what is happening. For a given equation, the computer applies Newton’s method for each starting value, the computer determines toward which root that value converges and assigns a color to the point. Shades of the color indicate how quickly that point comes to the root. The result is a fractal that often features large pools of color. These pools, known as basins of attraction, are “safe” areas where a starting point selected from there will come close to a root after a reasonable number of iteration.

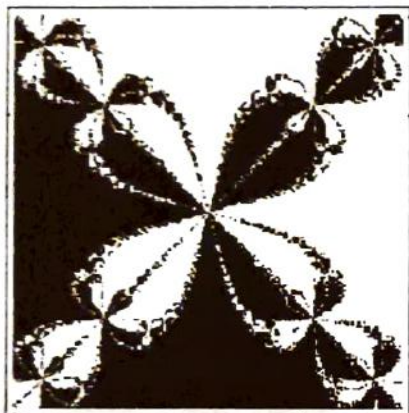


Figure 1. When Newton’s method is applied to the equation $z^4 - 1$, the resulting Julia set has four basins of attraction.

The visual results are intriguing. At the boundaries between the basins of attraction, all three basins and all three colors must meet at every point. That leads to an intricate network of color. Changing the parameter α shifts the boundaries and ruffles the basins of attraction. For certain values of the parameter, Newton’s method may take a point into a never-ending oscillation between two values, and never arrive at a root. In that case, the point falls into an attracting cycle. In some other cases, the basin of attraction around a zero practically disappears, and no choice of points leads to the root. The zeros themselves become unstable. The disappearance of basins of attraction marks the onset of chaos.

Other algebraic expressions such as $(z + \alpha)^{\alpha(z-1)}$ lead to equally startling and complicated patterns when Newton’s method is applied. The resulting pictures reveal intricate basins of attractions, evidence of bifurcations, and hints of chaos.

The boundaries, like the boundaries of the Mandelbrot set, are much more than simply dividing lines. They consist of elaborate and complex swirls and whirlpools that can pull Newton’s method into any one of the four roots. In these areas, a small shift in the starting point can lead to widely divergent results. And right on the convoluted boundary itself lie points that lead to no root. These points comprise the Julia set. Magnifying a section of the boundary region does not simplify the border. Further magnification merely reveals more miniature copies of the overall structure.

Mathematicians are also interested in determining which characteristics of a function lead to chaotic or unpredictable behavior. In 1879, Arthur Cayley, a British mathematician, explored the complexities of a

$$\begin{aligned} z_{n+1} &= z_n - \frac{f(z_1)}{f'(z_1)} \\ f'(z) &= 3z^2 - \alpha \\ z_{n+1} &= z_n - \frac{z^3 - \alpha z + a - 1}{3z^2 - \alpha} \\ &= \frac{z(3z^2 - \alpha) - (z^3 - \alpha z + a - 1)}{3z^2 - \alpha} \\ z_{n+1} &= \frac{2z^3 - \alpha + 1}{3z^2 - \alpha} \end{aligned}$$

The Pigeonhole Principle

Weikai Lang

The pigeonhole principle (a.k.a. the Dirichlet drawer principle, the shoebox principle) got its name from an example of pigeons flying into pigeonholes. In mathematical terms the principle states: *If $n+1$ pigeons are put into n pigeonholes, then at least one pigeonhole contains two or more pigeons.* This is known as the simple form of the pigeonhole principle.

The strong form is stated as follows: *Let $q_1, q_2, \dots, q_n$ be positive integers. If $q_1, q_2, \dots, q_n$ objects are put into n boxes, then either the first box contains at least q_1 objects, or the second box contains at least q_2 objects, ..., or the n th box contains at least q_n objects.*

We can see that the simple form of the pigeonhole principle results when $q_1=q_2=\dots=q_n=2$. Then we can see that: $q_1+q_2+\dots+q_n-n+1=2n-n+1=n+1$.

It follows that if we can prove the strong form of the pigeonhole principle, then the simple form is also true. The strong form can be proven indirectly as follows: Suppose we distributed $q_1+q_2+\dots+q_n-n+1$ objects among n boxes and for each $i=1, 2, \dots, n$, the i th box contained fewer than q_i objects. If we add the number of objects in all of the boxes together, $(q_1-1)+(q_2-1)+\dots+(q_n-1) = q_1+q_2+\dots+q_n-n$, we can see that the total number of objects in the boxes is one less than the total number of objects put in, which is $q_1+q_2+\dots+q_n-n+1$. Thus we can conclude that for some $i=1, 2, \dots, n$, the i th box contains at least q_i objects.

It is also true that if we put more than $q_1+q_2+\dots+q_n-n+1$ objects in n boxes, the strong form of the pigeonhole principle still holds true by mathematical induction.

If, in the strong form of the pigeonhole principle, the integers $q_1, q_2, \dots, q_n$ are all equal to some integer r , then another form of the principle is: *If $n(r-1)+1$ objects are put into n boxes, then at least one of the boxes contains r or more of the objects.*

Another form can be expressed as: *If n integers $m_1, m_2, \dots, m_n$ have an average $\frac{m_1+m_2+\dots+m_n}{n}$, which is greater than $r-1$, then at least one of the integers $m_1, m_2, \dots, m_n$ is at least r .*

To see how the two forms above relate to each other, we substitute $q_1, q_2, \dots, q_n$ for $n(r-1)+1$ (since they are both the total number of objects) and for $i=1, 2, \dots, n$ let m_i be the number of objects in the i th box. We can take the average to get: $\frac{m_1+m_2+\dots+m_n}{n} = \frac{n(r-1)+1}{n} = \frac{n(r-1)}{n} + \frac{1}{n} = (r-1) + \frac{1}{n}$

Since the average is greater than $r-1$, one of the integers m_i is at least r .

An application of this would be:

Suppose a cable company has just discovered a new way to bring its customers more channels and sharper pictures. The process involves two different cables, both having 200 filaments. Half of the filaments in each cable emit a beta charge and the other half a gamma charge.

Announcing the revolutionary process months ahead of schedule, supposedly to get a "step up" on the opposition (who, having superior foresight compared to your company's management, have already switched to digital broadcasts for its viewers), your company ordered two million of each cable. As the orders come in, it is apparent that something went awry with the orders. Only one type of cable is working the way it's supposed to and the other type was made with no stipulation on the number of beta and gamma filaments. There's no time to order new cables. As always, you are tasked with the burden of saving your company's reputation.

First, you tell your boss that there is no way of making the process work for all the customers all the time and that you'll try your best. Receiving no response other than a five-minute reprimanding, you set out on your impossible task. Suddenly, inspiration strikes you and you vaguely remember a section about pigeons sharing their holes or something. Anyway, you start thinking about this in a mathematical manner rather than just trying to connect the cables and hoping that you'll get lucky.

We know that there are 200 possible positions for the non-faulty cable (we shall now term this as *a*) to connect with the faulty cable (we shall now term this as *b*). Since the *a* cable has 100 of each filament, each filament in the *b* cable will connect correctly in exactly 100 of the 200 possible positions (assuming we want to connect beta to gamma or vice-versa). Thus the total number of correct connections over all the possible positions equals 20,000 (100×200). Therefore the average number of matches per position is 100, and there must be some position with at least 100 matches.

As your boss awaits your report, you march towards his office, pondering the combination of excuses which would prove the least painful in announcing that the company's \$36 million investment will work for all of its customers half the time...

Fermat's Last Theorem: The Basic Idea to the Solution

Zardosht Kasheff

In about 1637 (the exact time is unknown), Pierre de Fermat, a seventeenth-century French jurist and mathematician, wrote the following on the margin of his copy of Diophantus' Arithmetica, next to a problem on breaking down a squared number into two squares:

On the other hand, it is impossible to separate a cube into two cubes, or a biquadrate into two biquadrates, or generally any power except a square into two powers with the same exponent. I have discovered a truly marvelous proof of this, which, however, the margin is not large enough to contain.

What Fermat was saying was $x^n + y^n \neq z^n$ for all $n \geq 3$. This became known as Fermat's Last Theorem.

For over 340 years, this seemingly simple theorem remained unproven. Fermat himself proved the theorem for $n = 4$. In 1753, Leonhard Euler proved the theorem for $n = 3$. In 1828, Peter Dirichlet proved the theorem for $n = 5$. In 1839, Gabriel Lamé proved the theorem for $n = 7$. In 1857, Ernst Kummer proved the theorem for $n < 100$, and even as recently as 1992, with the assistance of computers, the theorem was proven for $n < 4,000,000$. Yet it was the general case that stymied all mathematicians.

In 1994, Andrew Wiles presented the world with a proof and mathematics' largest enigma was solved. However, Andrew Wiles did not construct the proof from ground zero. He built it upon the work of other mathematicians such as Henri Poincaré, Yutaka Taniyama, Goro Shimura, Gerhard Frey, Ken Ribet, Barry Mazur, and others.

The keys to Wiles' proof are modular forms and elliptic curves. Henri Poincaré studied periodic functions such as $\sin x$ and $\cos x$. A periodic function, according to the Dictionary of Science, is "a function that repeats the same values at regular intervals." What Poincaré studied were periodic functions in the complex plane. A periodic function in the complex plane has periodicity to both the real and imaginary axis. Hence, a periodic function in the complex plane has the property $f(z) = f((az+b)/(cz+d))$, the coefficients a, b, c, d are integers and $ad-bc = 1$. These periodic functions are called modular forms. Poincaré discovered that these modular forms are symmetric in many ways. These symmetries are obtained in the complex plane by adding a number to the function, and taking the reciprocal.

Elliptic curves started to become associated with Diophantine problems more and more in the twentieth century. An elliptic curve is the locus of points on the graph $y^2 = Ax^3 + Bx^2 + Cx + D$. Although elliptic curves may be easily defined, their usefulness in number theory and overall complexity is too extensive to explain in this paper. However, one important property about elliptic curves is that if one takes a rational point on the curve, and adds it with another rational point, the result will be a third point on the elliptic curve. The association between this, modular forms, and Fermat's Last Theorem was not seen until two mathematicians, Goro Shimura and Yutaka Taniyama, formulated the Taniyama-Shimura Conjecture.

Taniyama-Shimura Conjecture. *Given an elliptic curve $y^2 = Ax^3 + Bx^2 + Cx + D$ over $\mathbb{Q}$, there are non-constant modular functions $f(z), g(z)$ of the same level N such that $f(z)^2 = Ag(z)^3 + Bg(z)^2 + Cg(z) + D$.*

Hence, according to Taniyama and Shimura, every elliptic curve over $\mathbb{Q}$ has a “hyperbolic uniformization” by a modular form or in other words, is periodic. Therefore, every elliptic curve is modular.

The conjecture can be explained in simpler terms as follows: “If we fold the complex plane as a torus, then this surface will hold all solutions to elliptic equations over the rational numbers, these in turn arising from the equations of Diophantus [1].”

Wheat Gerhard Frey, a mathematician from the University of Heidelberg, realized was the first noticeable step to proving Fermat’s Last Theorem. He claimed that if the Taniyama-Shimura Conjecture could be proven true, it would lead to the proof of Fermat’s Last Theorem. Frey’s idea used contradiction to prove the theorem. He thought that if Fermat’s Last Theorem was false, then the solution a, b, c would arise to a certain elliptic curve. Frey’s equation of the elliptic curve was $y^2 = x(x - a^n)(x + b^n)$. This became known as Frey’s curve. Frey’s claim was that this curve was so weird, that it could not exist. Most important of all, the curve was not modular. According to the Taniyama-Shimura conjecture, all elliptic curves were modular. Therefore, if the Taniyama-Shimura conjecture is true, and this curve is not modular, then the curve does not exist, and therefore, no solutions to Fermat’s Last Theorem exist!

Frey’s reasoning was sound, but in his statement was only a conjecture. Not only did he substantiate his statement, nobody else had any proof either that the Taniyama-Shimura conjecture was true. Ken Ribet, a professor of mathematics at the University of California, took it upon himself to prove Frey’s statement. With some help from Barry Mazur, a mathematician at Harvard University, Ken Ribet proved Frey’s statement, but not the Taniyama-Shimura conjecture.

After Ribet’s proof, all that remained was to formulate a proof for the Taniyama-Shimura conjecture, and then Fermat’s Last Theorem would be proven. Andrew Wiles began to work on the conjecture, and, after seven years, proved it for semistable elliptic curves (only semistable elliptic curves were necessary for the proof).

Wiles’ basic approach was to “count” elliptic curves and “count” modular elliptic curves, and show that they are the same. The problem was that the set of elliptic and modular elliptic curves is infinite. What Wiles did was to try to transform elliptic curves into Galois representations and “count” them against Galois representations of modular elliptic curves. This transformation, named after its creator, Évariste Galois, takes infinite sets and transforms them to “groups with a finite set of elements [1].” Now Wiles was working with a finite set rather than an infinite set of numbers.

In 1993, Wiles presented a false proof of Fermat’s Last Theorem. Wiles had a two-hundred page proof which included everything from the work done by Frey to the Taniyama-Shimura conjecture. A flaw was found and Wiles had to use another approach. A year later, Wiles successfully proved Fermat’s Last Theorem using what is called the Horizontal Iwasawa Theory. The complete mathematics of the proof are many pages long and too complex to be summarized in this paper. (For further information in this, see [3]).

Wiles did successfully prove the Taniyama-Shimura conjecture, and Ribet did prove Frey’s statement. The proof was as follows: If Fermat’s Last Theorem is false, and there are solutions a, b, c, n , then the associated elliptic curve known as the Frey curve, $y^2 = x(x - a^n)(x + b^n)$, is non-modular and according to the Taniyama-Shimura conjecture, cannot exist.

Although Fermat’s Last Theorem has little practical functions in mathematics, what makes the theorem interesting and appealing is that in order to solve a seemingly simply theory, one was forced to use information from almost all fields of mathematics: geometry, topology (modular forms), number

theory (Galois representations) and simple applied algebra. Fermat's Last Theorem was truly mathematics' largest enigma.

References

- [1] Aczel, Amir D. Fermat's Last Theorem: Unlocking the Secret of an Ancient Mathematical Problem. New York, 1996.
- [2] Barnhart, Robert K. The American Heritage Dictionary of Science. Boston, 1986.
- [3] Coates, John and Yau, S.T. Elliptic Curves, Modular Forms, and Fermat's Last Theorem, Series in Number Theory, Vol. 1. Cambridge, 1995.
- [4] Cox, David A. "Introduction to Fermat's Last Theorem." American Mathematician Monthly, 101 (1994), No.1, pages 3-14.
- [5] Mahoney, Michael Sean. The Mathematical Career of Pierre de Fermat 1601-1665. Princeton, 1973.
- [6] Osserman, Robert. Fermat's Last Theorem: A Supplement to the Video. Berkeley, 1993.
- [7] Poorten, Alf van der. Notes on Fermat's Last Theorem. New York, 1996.

Continued Fractions

Jae Jang

Definition. A continued fraction is an expression of the form:

$$a_0 + \frac{b_1}{a_1 + \frac{b_2}{a_2 + \frac{b_3}{a_3 + \dots}}}$$

In the most general case, a_i and b_j can be either real or imaginary. Continued fractions with $b_j = 1$, $a_i \in \mathbb{N}$ for $i, j \geq 0$ are known as simple continued fractions. Since “simple” continued fractions alone have enough properties to fill a whole book, the best I can do here is to give a short introduction.

A brief history. The origin of continued fractions is not certain. Although used as far as 2000 years ago, their true groundwork was not established until the middle of the 17th century. The term “continued fraction” was coined by John Wallis (1616-1703), who published the first book on the topic in 1695, Opera Mathematica.

Finite simple continued fractions. Let us begin with a discussion of *finite* continued fractions (e.g. there are a finite number of a_i ’s). Usually, mathematicians express the simple continued fraction

$$a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \dots + \frac{1}{a_n}}}}$$

as $[a_0, a_1, a_2, \dots, a_n]$, where $a_1, a_2, \dots, a_n$ are the *partial quotients*, or simply the

quotients, of the continued fraction.

There are several methods to find the a_i ’s for a particular $\frac{p}{q}$. For example, the continued fraction representation of $\frac{31}{4}$ can be found by the following two ways. The most straightforward approach is to repeatedly simplify and manipulate the original fraction into the form of a simple continued fraction. We begin by rewriting the fraction into a mixed number, $\frac{31}{4} = 7\frac{3}{4}$. The result can be written as into $7 + \frac{3}{4}$ which equals $7 + \frac{1}{\frac{4}{3}} = 7 + \frac{1}{1 + \frac{1}{3}} = [7, 1, 3]$.

Another method uses Euclid's Algorithm, which employs the division algorithm: given a number $\frac{p}{q}$, where $p, q \in \mathbb{N}$, $\exists c_0$ such that $p = c_0 q + r_1$ and $0 \leq r_1 < c_0$. Also, $\exists c_1$ such that $q = c_1 r_1 + r_2$ and $0 \leq r_2 < c_1$. This process can be repeated until $r_n = 0$. The n c_i 's found are equal to a_i . So, to find $\frac{31}{4}$, $31 = 7 \times 4 + 3$; $4 = 1 \times 3 + 1$; $3 = 3 \times 1 + 0$.

Therefore, $\frac{31}{4} = [7, 1, 3]$. Although Euclid's algorithm is not practical for finding the continued fraction representations of small rational fractions, it is helpful when $\frac{p}{q}$ is large and irreducible. This algorithm can also be used to help prove that any positive rational $\frac{p}{q}$ can be represented by a simple continued fraction. Also, Euclid's algorithm can be extended to complex numbers and irrationals.

Definition. convergents

We call $[a_0, a_1, \dots, a_k]$, $0 \leq k \leq n$, the k^{th} partial convergent of the continued fraction. It can be seen that $[a_0, a_1, \dots, a_k] = a_0 + \frac{1}{[a_1, a_2, \dots, a_k]} = [a_0, [a_1, a_2, \dots, a_k]]$, $1 \leq k \leq n$. In general, $[a_0, a_1, \dots, a_k] = [[a_0, a_1, \dots, a_{m-1}], [a_m, a_{m+1}, \dots, a_k]]$, $1 \leq m < k \leq n$.

Theorems. Although simple in concept, continued fractions are rich in properties. Proofs of only the most elementary and important theorems are presented here.

Given a finite simple continued fraction $[a_0, a_1, \dots, a_n]$:

Theorem 1. If p_k and q_k are defined by $p_0 = a_0, p_1 = a_1 a_0 + 1, p_k = a_k p_{k-1} + p_{k-2}$ ($2 \leq k \leq n$) and $q_0 = 1, q_1 = a_1, q_k = a_k q_{k-1} + q_{k-2}$ ($2 \leq k \leq n$), then $[a_0, a_1, \dots, a_k] = \frac{p_k}{q_k}$, the k^{th} convergent.

Proof: Induction can be used to prove this. Obviously, the theorem holds for $k = 0$ and $k = 1$.

Assume it is true for all $k \leq m$, where $m < n$, $[a_0, a_1, \dots, a_{m-1}, a_m] = \frac{p_m}{q_m} = \frac{a_m p_{m-1} + p_{m-2}}{a_m q_{m-1} + q_{m-2}}$,

and prove that the theorem holds for $k = m + 1$. Using the definition of a convergent, we

$$\text{get } [a_0, a_1, \dots, a_{m-1}, a_m, a_{m+1}] = [a_0, a_1, \dots, a_{m-1}, a_m + \frac{1}{a_{m+1}}] = \frac{(a_m + \frac{1}{a_{m+1}})p_{m-1} + p_{m-2}}{\frac{1}{a_{m+1}}(a_m q_{m-1} + q_{m-2})} =$$

$$\frac{a_{m+1}(a_m p_{m-1} + p_{m-2}) + p_{m-1}}{a_{m+1}(a_m q_{m-1} + q_{m-2}) + q_{m-1}} = \frac{a_{m+1}p_m + p_{m-1}}{a_{m+1}q_m + q_{m-1}} = \frac{p_{m+1}}{q_{m+1}}. \text{ QED.}$$

Theorem 2. The functions p_k and q_k satisfy $p_k p_{k-1} - p_{k-1} q_k = (-1)^{k-1}$.

Proof: Substituting definitions of p_k and q_k , $p_k p_{k-1} - p_{k-1} q_k = (a_k p_{k-1} + p_{k-2}) q_{k-1} - p_{k-1} (a_k q_{k-1} + q_{k-2}) = -(p_{k-1} q_{k-2} - p_{k-2} q_{k-1})$. Repeat for the expression inside the parenthesis (which is

just the first expression with subscripts decreased by one). $p_k q_{k-1} - p_{k-1} q_k = (-1)^{k-1} (p_1 q_0 - p_0 q_1) = (-1)^{k-1}$. QED.

This theorem can be applied when solving linear Diophantine equations (refer to a book on number theory for more information on Diophantine equations).

Theorem 3. $p_k q_{k-2} - p_{k-2} q_k = (-1)^k a_k$.

Proof. The proof is similar to the proof for Theorem 2. $p_k q_{k-2} - p_{k-2} q_k = (a_k p_{k-1} + p_{k-2}) q_{k-2} - p_{k-2} (a_k q_{k-1} + q_{k-2}) = a_k (p_{k-1} q_{k-2} - p_{k-2} q_{k-1}) = (-1)^k a_k$. QED.

Theorems 2 and 3 are very useful when proving other theorems involving the relationships between convergents, such as:

Theorem 4. The even convergents $\frac{p_{2m}}{q_{2m}}$ increase strictly with m , while the odd convergents $\frac{p_{2m+1}}{q_{2m+1}}$ decrease strictly with m .

Proof. Divide the equation in Theorem 3 through by $q_k q_{k-1}$, to get $\frac{p_k}{q_k} - \frac{p_{k-2}}{q_{k-2}} = \frac{(-1)^k a_k}{q_k q_{k-2}}$. The expression on the left is the difference between the k^{th} convergent and the $(k-2)^{\text{th}}$ convergent. Since all a_i are positive in simple continued fractions, all q_i are also positive by definition. This means that the difference between the k^{th} convergent and the $(k-2)^{\text{th}}$ convergent always has the sign of $(-1)^k$. Therefore if k is even, then the difference between the k^{th} convergent and the $(k-2)^{\text{th}}$ convergent has the sign of $(-1)^{2m} = [(-1)^2]^m = 1$ (the positive sign indicates that the $(k-2)^{\text{th}}$ convergent is smaller than the k^{th} convergent), whereas if k is odd, then the difference between the k^{th} convergent and the $(k-2)^{\text{th}}$ convergent has the sign of $(-1)^{2m+1} = (-1)(-1)^{2m} = -1$ (the negative sign implies that the $(k-2)^{\text{th}}$ convergent is larger than the k^{th}). QED.

Try proving that $\frac{p_{2m+1}}{q_{2m+1}} > \frac{p_{2m}}{q_{2m}}$ using Theorem 2.

Theorem 5. The convergents of a simple continued fraction are in their lowest terms.

Proof. This can be proved easily by using Theorem 2. However, first notice that if $a, b, c \in \mathbb{Z}$, $a+b=c$ and $d \mid a$ (d divides a) and $d \mid b$, then $d \mid c$, the theorem can be proven using contradiction. Let $d \in \mathbb{Z}^+$ be the greatest common factor of p_k and q_k . Let us assume that $d \neq 1$ (assume that the fraction is not in lowest terms). Looking back at Theorem 2, observe that if $d \mid p_k$ and $d \mid q_k$ then d must divide $(-1)^{k-1}$. Since $(-1)^{k-1}$ can only be 1 or -1 , $d = 1$ (Contradiction). QED.

Theorem 6. If $a'_k = [a_k, a_{k+1}, \dots, a_n]$ (called the k^{th} complete quotient) then $\frac{p}{q} = a_0$, $\frac{p}{q} = \frac{a_1 a_0 + 1}{a_1}$, and

$$\frac{p}{q} = \frac{a_k p_{k-1} + p_{k-2}}{a_k q_{k-1} + q_{k-2}}, \quad (2 \leq k \leq n).$$

Proof:

Proof by induction: The two base cases are obviously true. Assume that the theorem is true for $k = m$ and prove that it is true for $k = m + 1$: $\frac{p}{q} = \frac{a_m q_{m-1} + p_{m-2}}{a_m q_{m-1} + q_{m-2}}$ (by induction

hypothesis), substitution for a_m yields $\frac{p}{q} = \frac{[a_m, \dots, a_n] p_{m-1} + p_{m-2}}{[a_m, \dots, a_n] q_{m-1} + q_{m-2}} =$

$$\frac{[a_m, [a_{m+1}, \dots, a_n]] p_{m-1} + p_{m-2}}{[a_m, [a_{m+1}, \dots, a_n]] q_{m-1} + q_{m-2}} = \frac{(a_m + \frac{1}{a_{m+1}}) p_{m-1} + p_{m-2}}{(a_m + \frac{1}{a_{m+1}}) q_{m-1} + q_{m-2}} = \frac{\frac{p_{m-1}}{a_{m+1}} + a_m p_{m-1} + p_{m-2}}{\frac{q_{m-1}}{a_{m+1}} + a_m q_{m-1} + q_{m-2}}.$$

$$\text{Substituting for the definitions of } p_m \text{ and } q_m, \frac{\frac{p_{m-1}}{a_{m+1}} + p_m}{\frac{q_{m-1}}{a_{m+1}} + q_m} = \frac{p_{m-1} + a_{m+1} p_m}{q_{m-1} + a_{m+1} q_m}. \text{ QED.}$$

Theorem 7. If $1 \leq k \leq n-1$ then $\frac{p}{q} - \frac{p_k}{q_k} = \frac{(-1)^k}{q_k q_{k+1}}$, where $q_{m+1} = a_{k+1} q_k + q_{k-1}$.

Proof: From Theorem 6, $\frac{p}{q} = \frac{a_{k+1} p_k + p_{k-1}}{a_{k+1} q_k + q_{k-1}}$, for $1 \leq k \leq n-1$. So, $\frac{p}{q} - \frac{p_k}{q_k} = -\frac{p_k q_{k-1} - p_{k-1} q_k}{q_k (a_{k+1} q_k + q_{k-1})}$.

Using Theorem 2, the expression becomes $-\frac{(-1)^{k-1}}{q_k q_{k+1}} = \frac{(-1)^k}{q_k q_{k+1}}$. QED.

Notice that Theorem 7 gives a value for the difference between a given fraction $\frac{p}{q}$ and its k^{th}

convergent. This property is just one of the various relationships between $\frac{p}{q}$ and its convergents.

There are numerous inequalities that allow one to find a convergent that is a close approximation of $\frac{p}{q}$ with given degree of error. General continued fractions can be utilized to find a convenient rational estimate for an irrational expression. Needless to say, this was just a very short introduction to continued fractions. Refer to any good book on number theory for a more comprehensive introduction.

Option Pricing Using the Black-Scholes Model

John Choi

Anybody who has ever played in a casino will have a very good feel for stock options. Basically, a stock option is a bet that the stock you are gambling on will come up a big winner or loser, depending what side of the fence you bet on. If the stock does not rise above your initial bet, then you lose. There is however, an opportunity to win huge profits. An option player has the potential to walk away with winnings that are many times his or her initial cash amount. Before going into some success stories, let's first define a few terms.

A stock option is comprised of two parts: a call option and a put option. A call option is for the optimist who believes the price of the current stock will go up. A call option contract gives the holder the right to buy 100 shares of the underlying stock at a set price any time during the life of the option, usually ranging from one to nine months. Put options are for pessimists who believe that the stock will fall in price. A put contract gives the holder the right to sell 100 shares of stock at a set price any time during the life of the option. The price at which you are allowed to buy the stock is known as the exercise or more commonly, the strike price. The cost of the option itself is referred to as the premium.

Occasionally, huge profits can be made, often reaching up to the ten thousands. Such stories of making 400% and above profits are not uncommon. It may surprise you that there is a pretty reliable way of figuring out which stock options will make you rich, though not to the extent of making you a millionaire. Nonetheless, it may very well get you a little more money than you have now. At any rate, there is a way of calculating theoretical prices on stock options that allow the investor to choose wisely.

If we limit our discussion to call options, then the value of a European call or an American call (a European call can only be exercised on its expiration date, while an American call can be exercised at any time during its life span) on a nondividend-paying stock is $C = SN(d_1) - Ke^{-rT}N(d_2)$ where S = the current price of the asset, K = the strike price of the call option, r = the risk-free interest rate, T = the time to expiration, $N(d)$ = the cumulative standard normal distribution function, $d_1 = \frac{\ln(S/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$, $d_2 = d_1 - \sigma\sqrt{T}$, σ = the annualized volatility of stock returns, specifically, the standard deviation of the asset's annual returns, $\ln(S/K)$ = the natural logarithm of S/K , and e^{-rT} = the exponential function of $-rT$. e^{-rT} is the present value factor when r is continuously compounded for T periods. Ke^{-rT} is the present value of K .

This formula is called the Black-Scholes Option Pricing Model. Created by Fischer Black and Myron Scholes, this formula has allowed investors to make true ballpark guesses on the theoretical value of their assets. However, certain assumptions were made in order for the formula to work as properly as possible.

1. Capital markets are perfect. There are no transaction costs or taxes, or short-selling constraints. Investors get full use of short-sale proceeds. All assets are infinitely divisible.
2. All investors can borrow and lend at the same riskless interest rate, which will be constant.
3. The stock pays no dividends. This means that the model can value European or American calls – since both will be exercised only at the expiration date – only if the call finishes in the money (the current stock price is higher than the strike price, allowing the investor to buy low and sell high).
4. Markets are always open and trading is continuous.

5. The stock price follows a specific type of stochastic process called a diffusion process. Since the mathematical base for diffusion processes is beyond the scope of this article, it is sufficient to state that diffusion processes are defined as following a certain behavior in stock returns. In a diffusion process, returns over a given interval of time are lognormally distributed whereas continuously compounded returns are normally distributed.

However, in reality most of these assumptions do not hold up well. For example, any investor knows that there is a commission charged for any type of transaction. Also, the riskless interest rate is always changing. To address this problem, many investors simply just take the interest rate of the nearest Treasury bill that expires around the time of the option. Nonetheless, according to Dan Bernstein, from the Options Department at Merrill Lynch, "[The Black-Scholes Option Pricing Model] will still give you ballpark figures, something that was previously unheard of." Prior to the fruition of the Black-Scholes model, option valuation would usually be laced with psychological undercurrents. Psychologists were just as, or even more credible, as options analysts!

To prove the ease of using the Black-Scholes model, let's try an example:

Let's say that a stock of the NYNEX Corp. is selling for $58\frac{7}{8}$. The riskless interest rate is 8% per year. Estimate the value of the call with a strike price of 60 and a time to expiration of 3 months. The standard deviation of the stock's annual return is 0.22% per year.

Given this information, we have $S = 58.875$, $K = 60$, $T = 0.25$ years, $r = 0.08\%$ per year, and $\sigma = 0.22\%$ per year. Furthermore, $d_1 = \frac{\ln(S/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}} = \frac{\ln(58.875/60) + (0.08 + 0.0484/2)0.25}{0.22\sqrt{0.25}} =$

0.0647 , and $d_2 = d_1 - \sigma\sqrt{T} = 0.0647 - 0.22\sqrt{0.25} = -0.0453$. The table of the cumulative standard normal distribution functions, which can be found in reference [1], determines $N(d_1)$ and $N(d_2)$; $N(d_1) = 0.5258$ and $N(d_2) = 0.4819$. Finally, $Ke^{-rT} = 60 e^{-(0.08)(0.25)} = 58.8199$, so $C = SN(d_1) - Ke^{-rT}N(d_2) = 58.875(0.5258) - 58.8199(0.4819) = 2.6150$.

In actuality, most of the data for this example came from a Merck, a pharmaceutical company, call option. For this example, the value of σ had been changed in order for the model to give a fairly accurate theoretical value. It should be noted that the Black-Scholes Model is not dependent on the expected return on the stock, but rather on the annual volatility of the returns. The reason why taking the volatility is a better approach is because the volatility takes into account the probability that the returns may vary.

One may ask, "how are the values for the formula determined?" Well, the values of S , K and T can be found in any respectable newspaper (e.g. The New York Times, The Wall Street Journal), the risk free interest rate can be estimated by taking the rate of the T-bill whose expiration date is near the date of the option's expiration date, $N(d)$ can be looked up in [1], and the volatility can be found on the internet at sites such as "The Chicago Board of Options Home Page".

Now that we know the ease of use of the formula, the next question is how can we better understand it? Well, we must first introduce *delta*, δ , to represent the hedge ratio. *Delta*, as we all know, is the Greek letter which represents change. The hedge ratio is the theoretical change in the theoretical value of an option in relation to the change in the price of the stock. For example, if a call has a hedge ratio of 40, the call should increase 40 percent of the change in the stock if the stock rises, or decrease 40 percent of the price change if it decreases. If the price of the stock rises 1 point, the theoretical value of the call will increase 0.4 of a point, and decrease by 0.4 of a point if the stock drops 1 point. The value of the hedge ratio is usually good for the first one point move in the stock over a short period of time. An option with a high hedge ratio is usually a better purchase, since the value of the option will have a greater percentage gain (or loss) with the movement of the stock price. Options

that have a high hedge ratio value are usually in-the-money, and therefore more expensive, while options that have a low hedge ratio value are usually out-of-the-money. We will explore how the theoretical value and hedge ratio change as we toggle the one of the five parameters (stock price, expiration date, annual dividend, riskless interest rate and volatility) while keeping the others constant.

Taking the conclusions of our observations, we can sum up everything using variables. Taking the partial derivative of the Black-Scholes formula, $C = SN(d_1) - Ke^{-rT}N(d_2)$, will yield these five values:

$$\frac{\partial C}{\partial S} = N(d_1), \quad \frac{\partial \Delta}{\partial S} = \frac{N'(d_1)}{S\sigma\sqrt{T}}, \quad \frac{\partial C}{\partial \sigma} = S\sqrt{T}N'(d_1), \quad \frac{\partial C}{\partial T} = Ke^{-rT}[\sigma N'(d_2)/2\sqrt{T} + rN(d_2)], \quad \text{and}$$

$$\frac{\partial C}{\partial r} = TKe^{-rT}N(d_2).$$

To make things simpler, we can designate each parameter a specific symbol:

$$\frac{\partial C}{\partial S} = \Delta \text{ (Delta)}, \quad \frac{\partial \Delta}{\partial S} = \Gamma \text{ (Gamma)}, \quad \frac{\partial C}{\partial \sigma} = v \text{ (Vega)}, \quad \frac{\partial C}{\partial T} = \Theta \text{ (Theta)}, \quad \frac{\partial C}{\partial r} = \rho \text{ (Rho)}.$$

With these five parameters, we can calculate how much an option's price will fluctuate. *Delta* (Δ), *theta* (Θ), *rho* (ρ) and *vega* (v) can be dealt with cleanly through partial derivatives. *Gamma* (Γ) is the only parameter that needs a bit of explaining. *Gamma*, the change in delta over the change in stock price, brings the meaning of *delta* to a new dimension. For the open-minded investor, it is not just enough to know the current state of *delta*, but rather, the *trend* of *delta* is just as important.

It should be noted that if S increases, then C and *delta* also increase, and vice-versa if S decreases. Interest rates will also share the direct relationship with C . Also, as time increases, the values of C and *delta* also go down. As volatility increases, the values of C and *delta* rise, while their values decrease when volatility decreases. On the other hand, the values of C and *delta* decrease when annual dividends are increased while they increase when the annual dividends are lowered.

With all this knowledge of how stock option prices respond to individual minute changes, we must put everything into perspective. Multivariable changes must be taken into consideration. For example, let's say that AT&T decides to raise its annual dividend, which decreases the theoretical value. However, in the bigger picture, the stock price usually goes up, since investors see that they will get a better share of profits when AT&T raises the amount of money (or profits) that is given out to shareholders per share. So, when the stock price goes up, this sets off a probable lowering of volatility. If the AT&T stock takes a sharp turn (for better or worse) the volatility will reflect that.

The Black-Scholes Option Pricing Model was originally modeled after European options, where the investor could only exercise the option at the end of its expiration date, but can be applied to American options (where it can be exercised at any time), with a few minor adjustments. In fact, the Black-Scholes Option Pricing Model can be applied to any option. This includes futures options (dealing in commodities, like grain and livestock), index options (predicting the behavior of certain market indicators, e.g., Dow Jones Industrials, NASDAQ 100, the Japanese Nikkei 225), and currency indices (the different currencies of the world, e.g. the Korean Won, the Japanese Yen, or the U.S. dollar). Although out of date, the Black-Scholes Model still serves as the foundation for many computer models dealing with stock options.

References

- [1] Dubofsky, David A. Options and Financial Futures: Valuation and Uses. McGraw-Hill, Inc.: New York, 1992
- [2] Fullman, Scott H. Options: A Personal Seminar. Simon & Schuster: New York, 1992
- [3] Chriss, Neil A. Black-Scholes and Beyond. Times Mirror Professional Publishing Ltd.: New York, 1997

Special Magic Squares

Sung Hoon Kim

At some point during elementary school, everyone has encountered magic squares, a square organization of numbers, such that each row, column, and diagonal adds up to the same sum, the magic number. By investigating further into the subject of magic squares, I stumbled across certain arrangements which demonstrated special characteristics. I thought it might be fun to share some of these special magic squares which display amazing characteristics.

The first magic square is the Lo Shu, which is a basic 3×3 square. The magic number is 15, like all other third order squares. Why is this simple magic square special? Well, it is said to be the very first magic square that was discovered. Legend claims that it was spotted on the back of a tortoise by a Chinese Emperor. People of that time felt that this special arrangement of numbers possessed divine powers. This belief later led to the use of magic squares as instruments of magic and good luck talisman.

4	9	2
3	5	7
8	1	6

This next magic square is credited to Benjamin Franklin. The 8×8 magic square shown here adheres to the normal properties of an eighth order square: the sum of the rows and columns is 260 (the magic sum). When the first four numbers in any row or column are added together, the sum is 130, which is one half the magic number. Taking the sum of any four numbers equidistant from the center (such as 52, 45, 16, 17 or 5, 57, 40, 28) will likewise result in 130. Finally, the four numbers that form any smaller 2×2 square (such as 52, 61, 3, 14) also add up to 130.

52	61	4	13	20	29	36	45
14	3	62	51	46	35	30	19
53	60	5	12	21	28	37	44
11	6	59	54	43	38	27	22
55	58	7	10	23	26	39	42
9	8	57	56	41	40	25	24
50	63	2	15	18	31	34	47
16	1	64	49	48	33	32	17

The third magic square is very similar to the square of Benjamin Franklin. It is also a valid eighth order magic square, and possesses the property where the first four elements of any row or column sums to 130. However, that is not what makes this magic square special. If you start from the square labeled '1', you will see that you can trace all sixty-four numbers in

1	48	31	50	33	16	63	18
30	51	46	3	62	19	14	35
47	2	49	32	15	34	17	64
52	29	4	45	20	61	36	13
5	44	25	56	9	40	21	60
28	53	8	41	24	57	12	37
43	6	55	26	39	10	59	22
54	27	42	7	58	23	38	11

succession by using "knight jumps" (such as those used in the game of chess). This incredible magic square was created in the beginning of the 18th century.

Finally, the fourth magic square is special in that it is created from only prime numbers. This ninth order magic square has a magic sum of 163,953 (feel free to test this out for yourself). While this magic square uses many large prime numbers, it does not hold any other special property.

18199	20533	15901	18229	19417	17293	17827	29983	6571
15913	18211	20509	17377	18313	19249	6871	18127	29383
20521	15889	18223	19333	17209	18397	29683	6271	18247
18097	20479	15823	17851	33073	3727	18289	36583	31
15859	18133	20407	4093	18217	32341	41	18301	36559
20443	15787	18169	32707	3361	18583	36571	19	18313
17977	32917	4027	18061	21649	14653	18013	31543	5113
4357	18307	32257	14713	18121	21529	5323	18223	31123
32587	3697	18637	21589	14593	18181	31333	4903	18433

Here I have presented four of the many special magic squares that exist in the mathematical world. Through a small amount of research, I have been able to find these special squares, as well as several others, all of which possess certain characteristics. There is a whole wide world of magic squares, ready to be explored. Perhaps you'll be able to discover your own magic square that displays its own special characteristics that will amaze others.

An Interview with Daniel Stronger

The "BORIS"

In 1997, Stuyvesant High School's Daniel Stronger received the nation's second-highest score on the USA Math Olympiad, which qualified him to represent the United States on a six-person team that competed in the International Math Olympiad (IMO) in Buenos-Aires, Argentina, in late July. Before the IMO, Daniel was one of thirty students from across America who attended a four-week Math Olympiad Program (MOP) in the summer at the University of Nebraska at Lincoln. Daniel, who was a captain of the Stuyvesant Math Team for two years and is currently a freshman at Harvard University, was asked several questions about his experiences at MOP and at the IMO in 1997.

How was MOP organized? Where did you live? What was the daily schedule? What did you learn? What types of people were there? Were there any particularly fun incidents that happened?

MOP took place over four weeks each summer at the University of Nebraska at Lincoln campus. We lived in the college rooms that weren't being used over the summer. It always started painfully early, forcing me to miss finals week and Regents. The idea behind MOP was to prepare our IMO team for the IMO, and to prepare other non-seniors to potentially compete at future IMOs. In an attempt to do this, on weekdays, we would have three hours of class in the morning, in which they would teach us problem-solving techniques. In the afternoons, three days a week, we would have MOP-tests, which were about three and a half-hours long for three problems. I found these to be the most useful for getting better at solving problems. By far the best way to improve at anything is to try it and do poorly. I can assure you there were problems on those that we all had no idea how to do. These, of course, were the most useful ones. Learning how to do a problem that I struggled with helplessly for hours is often frustrating, but usually very enlightening. The following is my favorite MOP-test problem: *Take any five points on a circle. For a given pair of the five points, one can take the line through those two, and the centroid of the triangle formed by the other three, and from those, form the line through that centroid perpendicular to that line. Since there are ten possible initial pairs of points, there are ten lines that can be formed like that. Show that all ten of them are concurrent.* ☺

MOP is officially the Math Olympiad Summer Program, abbreviated by the administration as "MOSP." The students hate the 'S' with a vengeance. MOP is obviously a cooler name, and that is what we will call it, and we will be MOPpers. This conflict epitomizes a more general conflict between the students and the administration of MOP. The students try to keep it cool and enjoy it, do math, and have fun. The administration tries to impose rules, make us do homework, and make us go to bed. We win.

MOP is tremendously fun. We have enough free time to have fun. For some reason, MOPpers are eternally obsessed with ultimate frisbee. It is our sport. Over my three summers there, I've become awesome at ultimate. I love it too. Personally, I think math people like ultimate because the skill of figuring out where the frisbee is going to go has some analytical aspect to it. I haven't found anyone who agrees with me (they all claim that it's just traditional or something). Aside from frisbee, we do lots of other cool stuff. In our free time (of which there is much), we hang out, watch movies, play weird games, normal games, etc. We have fun. Every now and then, we work on math too, but it's cool; it's as uncompetitive as you could expect it to be. Working on math with MOPpers can be anything from nice and fun to unforgettable and awe-inspiring.

The best part about MOP is the MOPpers. I mean, if you take the 30 or so best math kids in the country or so, and lock them all together for a month, you have to realize you'll be getting some weird kids (mostly in a good way). I don't know why, but when you get that far off the end of the bell curve, people's other attributes start getting skewed too. Everyone there is weird in his or her own way. They vary from nasty freaks to cool freaks to just awesome people. I love MOPpers; I just can't describe how uniquely amazing they are. Once you weed out the nasty freaks, you're left with some really cool kids. I particularly liked them because they gave me the unique opportunity to talk to people who I could relate to, who had also spent their childhood starving for people who could never understand what they were talking about. We needed each other. This is beginning to be way too reminiscent of my college essay. All right, all right – I confess: I'm having trouble making my point. The point is MOPpers are cool, okay?

How did you do as a team at IMO? How did you do as an individual? What was your schedule? How much fun did you get to have? Also, tell us about the time you got drunk. ☺

In 1997, the United States tied for fourth at the IMO. The countries that beat us were China, Hungary, and Iran (in that order), and we tied Russia. The results usually look something like that, with the US around fourth or so, China at first or second. The others in the top ten were Ukraine, Bulgaria, Romania, Australia, and Vietnam.

Each of the six problems was worth seven points (partial credit given, but not very generously). Each team consisted of six people. Our score was 202 out of 252. My score was 29 out of 42, which I was rather disappointed with, and tied for last among our team (our scores were 29, 29, 30, 32, 40, and 42). My score put me in a big tie for about 70th place worldwide, and qualified me for a silver medal, of which I am extremely proud. I like to be a jerk and look to see which countries I beat all by myself. For example, Chile's total score was 28; or more sadly, Guatemala's six people got a total of 7. ☺

Fun IMO stuff: take 500 of the best math kids from all around the world, and stick them all together in a big hotel in Mar del Plata (just south of Buenos Aires), give them some money, access to the Argentinean streets, and a big game room with lots of toys and video games and an air hockey table for a couple of weeks and see what happens. It was cool. The six of us from the US team (Josh Nichols-Barrer, John Clyde, Li-Chung Chen, Carl Bosley, Nathan Curtis, and I) stayed there for a total of a couple of weeks, and the test was only over a couple of days. We spent the time having fun, listening to good music, playing air hockey, whatever we wanted. We had computer access, etc. They forced us to go on some "interesting outings," but they were generally boring, but hey – what are you going to do? We had fun anyway. We played our traditional ultimate game against the Canadians (and beat them, as of course the tradition dictated), so that was cool. We became pretty good friends with the UK team; we played cards with them and stuff. They were cool. "Hey, they really do say 'Bloody hell!'" (-John Clyde). Hey, have you ever played around the world ping-pong with the Colombian IMO team? It's fun.

Each team was given an Argentinean person who was familiar with Argentina and who also knew the team's language. How they found an Argentinean who spoke Icelandic, I'll never know. Our guide's name was Ariel, and he ruled. All right, all right, you're all skimming to my story of when I got drunk? Here it is: one night (after the IMO), Ariel mentioned to us that if we wanted to, he could take us to a local bar and get us some drinks. I had never been drunk before (or been out of the country, but that's not quite as relevant), so I said, "what the hell, let's go." John is from Idaho so he was content with an opportunity to get drunk. Josh came along to not drink too much and make sure we didn't do anything too stupid. John had some hi-tech drink whose name apparently translated to 'gasoline,' so I decided to go with a banana daiquiri instead. It was very good, and I'm sorry to disappoint you, but I

drank sufficiently little that my level of drunkenness was about half (I'd say I was half drunk), and it was really cool. John got much more drunk, but neither of us got sick at all. Being half drunk was fun. I'd say of all getting drunk experiences, mine was the most awesome. I had fun. John was very funny drunk. I have a memory of him sitting on the floor thinking, watching us play cards with the UK team, and periodically saying completely senseless things about the hardest IMO problems. It was very amusing. Nonetheless, at the same time, the whole thing bothered me. In the end, I decided the lack of inhibitions and playfulness that come from being drunk weren't worth the loss of reasoning ability and ability to make decisions (hey, I have a lot of respect for my reasoning ability and I don't like the idea of throwing it out the window every now and then just for the hell of it – there are better fun things to do, always), and I resolved not to drink again. I've managed to not touch any alcohol during my first term at Harvard, although admittedly this is not much of an accomplishment since there is very little pressure to drink here, for which I am thankful.

IMO girls? Okay, okay, I have to admit, there weren't very many (maybe a tenth of the contestants were girls), but there were some very cute ones. Josh and I particularly found cute this Bosnian girl, and though we somehow determined that she probably spoke English, we never really got around to talking to her. However, some of the female guides were nice. The Swedish guide lived next door to us, and John liked her. I have to admit we were childish – we spent hours one day (they forced us to go on a trip to the zoo that was boring as hell) trying to take pictures of the UK guide without her noticing. We have some funny pictures. It was cool. Honestly though, the cute guides were the Chinese and Hong Kong ones (although they spoke very little English, I was able to communicate with them through Li-Chung). Anyway, here's food for thought: there hasn't been a girl on the US IMO team ever; what do you think about that?

Math Survey AP Calculus Review

Originally by David Z. Weinstein, updated by Math Survey Staff

Asterisks (*) indicate material that applies only to BC Calculus

Limits

- Def'n: The limit of a function at a point is the value the function approaches at that point. That is, for $\lim_{x \rightarrow c} f(x) = L$, as x approaches c , the limit of $f(x)$ is L .
- Limits may fail to exist if:
 - $\lim_{x \rightarrow c^-} f(x) \neq \lim_{x \rightarrow c^+} f(x)$ (e.g. $\lim_{x \rightarrow 0} \frac{|x|}{x}$).
 - values of $f(x)$ increase without bound (e.g. infinite limits $\lim_{x \rightarrow 0} \frac{1}{x^2}$).
 - values of $f(x)$ oscillate indefinitely, approaching no limit (e.g. $\lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right)$).
- Basic limit theorems:
 - $\lim [f(x) \pm g(x)] = \lim f(x) \pm \lim g(x)$.
 - $\lim [f(x) \cdot g(x)] = \lim f(x) \cdot \lim g(x)$.
 - $\lim [f(x)/g(x)] = \lim f(x)/\lim g(x)$ if $\lim g(x) \neq 0$.
 - $\lim \sqrt[n]{f(x)} = \sqrt[n]{\lim f(x)}$.

Continuity

- Def'n: A function f is continuous at point a if $f(a)$ is defined, $\lim_{x \rightarrow a} f(x)$ exists, and $\lim_{x \rightarrow a} f(x) = f(a)$.
- Basic continuity theorems:
 - If f and g are continuous functions, then $f \cdot g$, cf , $f \pm g$, and f/g (for $g(x) \neq 0$) are also continuous.
 - All polynomial functions are continuous.
 - If a function f is continuous on a closed interval $[a, b]$, then the following are true:
 - Extreme Value Theorem: $f(x)$ has a minimum and maximum value on the interval.
 - Intermediate Value Theorem: If $f(a) < k < f(b)$, then there exists at least one value of c (where $c \in [a, b]$) such that $f(c) = k$. In other words, $f(x)$ cannot pass from one value to another in the interval without taking every intermediate value.
- Continuity does not imply differentiability!

Derivatives

- Def'n: Let $x = c$ be a point on the curve $y = f(x)$. Where $\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$ exists, the function is said to be differentiable. The limit is called the derivative of y with respect to x and is

denoted by $\frac{dy}{dx}$, $f'(x)$, $D_x y$, or y' . Furthermore, at a point a , the derivative is equal to the slope of the

tangent line to the curve at a , or $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$.

a. Second derivative: $d^2 y / dx^2 = f''(x) = D_x^2 y = y''$.

b. Third derivative: $d^3 y / dx^3 = f'''(x) = D_x^3 y = y'''$.

2. Table of derivatives (c, n are constants; u, v are differentiable functions of x):

$$\frac{d(c)}{dx} = 0$$

$$\frac{d(cu)}{dx} = c \frac{du}{dx}$$

$$\frac{d(u^n)}{dx} = nu^{n-1} \frac{du}{dx}$$

$$\frac{d(u+v)}{dx} = \frac{du}{dx} + \frac{dv}{dx}$$

$$\frac{d(u \cdot v)}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$\frac{d\left(\frac{u}{v}\right)}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$\frac{d(\sin u)}{dx} = \cos u \frac{du}{dx}$$

$$\frac{d(\cos u)}{dx} = -\sin u \frac{du}{dx}$$

$$\frac{d(\tan u)}{dx} = \sec^2 u \frac{du}{dx}$$

$$\frac{d(\cot u)}{dx} = -\csc^2 u \frac{du}{dx}$$

$$\frac{d(\sec u)}{dx} = \sec u \tan u \frac{du}{dx}$$

$$\frac{d(\csc u)}{dx} = -\csc u \cot u \frac{du}{dx}$$

$$\frac{d(\ln u)}{dx} = \frac{1}{u} \frac{du}{dx}$$

$$\frac{d(\log_c u)}{dx} = \frac{1}{\ln c} \cdot \frac{1}{u} \frac{du}{dx}$$

$$\frac{d(e^u)}{dx} = e^u \frac{du}{dx}$$

$$\frac{d(c^u)}{dx} = \ln c \cdot c^u \frac{du}{dx}$$

$$\frac{d(\arcsin u)}{dx} = \frac{1}{\sqrt{1-u^2}} \frac{du}{dx}$$

$$\frac{d(\arccos u)}{dx} = \frac{-1}{\sqrt{1-u^2}} \frac{du}{dx}$$

$$\frac{d(\arctan u)}{dx} = \frac{1}{1+u^2} \frac{du}{dx}$$

$$\frac{d(\operatorname{arccot} u)}{dx} = \frac{-1}{1+u^2} \frac{du}{dx}$$

$$\frac{d(\operatorname{arcsec} u)}{dx} = \frac{1}{\sqrt{u^2-1}} \frac{du}{dx}$$

$$\frac{d(\operatorname{arccsc} u)}{dx} = \frac{-1}{\sqrt{u^2-1}} \frac{du}{dx}$$

3. Chain rule: If $y = f(u)$ and $u = g(x)$ ($y = f(g(x))$), then $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = f'(u)g'(x) = f'(g(x))g'(x)$.

4. Inverse derivatives: If $y = f(x)$ is single-valued and has nonzero derivatives on a given interval, then the inverse function $f^{-1}(x)$ exists and is differentiable. Suppose $y = f(x)$ has an inverse function that is differentiable, then $f^{-1}(y)$ is $\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}}$.

5. * Parametric functions: If $y = f(t)$ and $x = g(t)$, then

a. $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$

b. $\frac{d^2 y}{dx^2} = \frac{dy'}{dx} \cdot \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}}$

c. $\frac{d^3 y}{dx^3} = \frac{dy''}{dx} \cdot \frac{\frac{d}{dt}\left(\frac{d^2 y}{dx^2}\right)}{\frac{dx}{dt}}$

6. Logarithmic differentiation: If $y = x^{g(x)}$, then $\ln y = \ln x^{g(x)} = g(x) \ln x$. Differentiating yields $\frac{1}{y} \frac{dy}{dx} = g'(x) \ln x + g(x) \frac{1}{x}$ or $\frac{dy}{dx} = y \left(g'(x) \ln x + g(x) \frac{1}{x} \right)$.

7. Note that differentiability implies continuity, but not vice versa (e.g., $f(x) = |x|$ at $x = 0$ is continuous but not differentiable).

Curve Sketching

1. Without Derivatives

- Intercepts: To get the x intercepts, set $y = 0$ and solve for x ; to get the y intercepts, set $x = 0$ and solve for y .
- Symmetry: If a curve has form $y = F(x)$ or $f(x, y) = 0$, then it is:
 - Symmetric to the x -axis if $f(x, y) = f(x, -y)$, or $F(x) = -F(x)$.
 - Symmetric to the y -axis if $f(x, y) = f(-x, y)$, or $F(x) = F(-x)$.
 - Symmetric to the origin if $f(x, y) = f(-x, -y)$, or $F(x) = -F(-x)$.
- Domain and range: Find x (domain) and y (range) values on which the function is defined.
- Asymptotes:
 - Vertical asymptotes: Let $y \rightarrow \pm\infty$ and find the limiting value(s) of x .
 - Horizontal asymptotes: Let $x \rightarrow \pm\infty$ and find the limiting value(s) of y .
 - We can also solve for one variable in terms of the other; set denominator equal to 0, etc.

2. With Derivatives

- Def'n:
 - Given x_1 and x_2 ($x_2 > x_1$) that belong to an interval of function $f(x)$, then the function is increasing where $f(x_2) > f(x_1)$ and decreasing where $f(x_2) < f(x_1)$.
 - $f(x)$ is a smooth function if both $f(x)$ and $f'(x)$ are continuous.
 - Critical values of $f(x)$ are values where $f'(x) = 0$ or is undefined.
 - A smooth curve is concave upward at a point if successive tangents to the curve from that point turn counterclockwise (slope increases – curve lies above tangent). A smooth curve is concave downward at a point if successive tangents to the curve from that point turn clockwise (slope decreases – curve lies below tangent).
- Important Theorems:
 - Function $f(x)$ is increasing at $x = a$ if $f'(x) > 0$, decreasing at $x = a$ if $f'(x) < 0$.
 - If $f(x)$, a continuous, differentiable function, has a relative extrema at $x = a$, then $f'(x) = 0$ (but not conversely).
 - Tests for Maxima and Minima
 - A function f has a relative maximum at point a if $f'(a) = 0$ and $f''(a) < 0$ (concave down). In other words, $f'(x)$ changes from positive to negative at $x = a$.
 - A function f has a relative minimum at point a if $f'(a) = 0$ and $f''(a) > 0$ (concave up). In other words, $f'(x)$ changes from negative to positive at $x = a$.
 - Concavity
 - A curve is concave up at point a if $f''(a) > 0$.
 - A curve is concave down at point a if $f''(a) < 0$.
 - Concavity changes at points of inflection (when $f''(a) = 0$ or is undefined).
- Absolute extrema can be attained where $f'(x) = 0$, at the endpoints of an interval, and where $f'(x)$ does not exist.

Applications of the Derivative

- If a function is differentiable at a point, then it must be continuous at the point, but not conversely.
- Rolle's Theorem: If $f(x)$ is continuous on $[a, b]$, $f'(x)$ exists on (a, b) , and $f(a) = f(b) = 0$, then there exists point c , where $a < c < b$, such that $f'(c) = 0$.
- Mean Value Theorem for Derivatives: If $f(x)$ is continuous on $[a, b]$ and $f'(x)$ exists on (a, b) , then there exists a point c , where $a < c < b$, such that $f'(c) = \frac{f(b) - f(a)}{b - a}$.

4. L'Hôpital's Rule

- a. If $f(x)$ and $g(x)$ are continuous and differentiable on an interval including point a , and

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{0}{0} \text{ or } \frac{\infty}{\infty}, \text{ then } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}.$$

- b. Evaluating other indeterminate forms:

- $0 \cdot \infty$. Change to $(1/\infty) \cdot \infty$ and use L'Hôpital's Rule.
- $\infty \cdot \infty$. Transform by using algebraic or trigonometric manipulation to acceptable form.
- 1^∞ , 0^0 , and ∞^0 . Take the log of the function and change to one of the above forms.

5. Differentials. Def'n: When $y = f(x)$, the differential of x (independent value) $dx = \Delta x$ and the differential of y (dependent variable) $dy = f'(x) dx$.

Def'n. For $y = f(x)$, the differential of x , dx , is approximately equal to Δx . The differential of y is $dy = f'(x)dx$. Note: For small Δx , $dy \approx \Delta y$, and $f'(x) \approx \Delta y/\Delta x$.

6. Related rate problems: Find the relationship between the variables, take the derivative with respect to time, and substitute.

Motion of Particles

1. Rectilinear motion (motion along a straight line)

- If the position of the particle is represented by $s = f(t)$ at time t , then
 - Average velocity: $v_{av} = \Delta s/\Delta t = (\text{total distance traveled})/(\text{time elapsed})$.
 - Average acceleration: $a_{av} = \Delta v/\Delta t = (\text{change in velocity})/(\text{time elapsed})$.
- Instantaneous velocity and acceleration for $s = f(t)$ at time t ,
 - Instantaneous velocity: $v = ds/dt = f'(t)$. A particle is instantaneously at rest if $v = 0$.
 - Instantaneous acceleration: $a = dv/dt = d^2s/dt^2 = f''(t)$. If a and v are of the same sign, then the speed of the particle $|v|$ is increasing; if a and v are of opposite sign, then $|v|$ is decreasing.

- 1.* Curvilinear motion (motion along a curve)

- For parametric equations of position $x = f(t)$ and $y = g(t)$:
 - In the x direction, the component of v is $v_x = dx/dt$ and the component of a is $a_x = dv_x/dt$.
 - In the y direction, the component of v is $v_y = dy/dt$ and the component of a is $a_y = dv_y/dt$.

$$\text{iii. } \|v\| = \sqrt{v_x^2 + v_y^2}, \tan \theta = \frac{v_y}{v_x} \text{ (quadrant is dependent on the signs of } v_x \text{ and } v_y).$$

$$\text{ii. } \|a\| = \sqrt{a_x^2 + a_y^2}, \tan \alpha = \frac{a_y}{a_x} \text{ (quadrant is dependent on the signs of } a_x \text{ and } a_y).$$

$$\text{b. Since } \tan \theta = \frac{dy}{dx} = \frac{dy/dt}{dx/dt} \text{ and } v^2 = v_x^2 + v_y^2 = \left(\frac{ds}{dt}\right)^2 = \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2, \|v\| = \left|\frac{ds}{dt}\right|. \text{ It follows that}$$

the velocity vector is tangent to the curve and the speed of the particle is $\left|\frac{ds}{dt}\right|$, where s is the distance along the curve from some fixed point.

Integration

1. Indefinite integral. Def'n: $\int f(x)dx = F(x) + C$ if $F'(x) = f(x)$.

2. Definite integral. Def'n: If the limit of the Riemann sum of $f(x)$ exists on $[a, b]$, then we say that f is integrable on $[a, b]$ and we denote the limit by $\lim_{\|\Delta\| \rightarrow 0} \sum_{i=1}^n f(c_i) \Delta x_i = \int_a^b f(x) dx$.

Properties:

- $\int_a^a f(x) dx = 0$.
 - $\int_a^b f(x) dx = - \int_b^a f(x) dx$.
 - $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$ for any c .
 - $\int_a^b k dx = k(b-a)$.
 - $\int_a^b (u+v) dx = \int_a^b u dx + \int_a^b v dx$.
3. Riemann sums:
- For a function f defined on $[a, b]$, where $[a, b]$ is divided into n partitions (such that the length of the longest partition is called the norm and is denoted by $\|\Delta\|$), the Riemann sum is $\sum_{i=1}^n f(c_i) \Delta x_i$, and the area under f is equal to $\lim_{\|\Delta\| \rightarrow 0} \sum_{i=1}^n f(c_i) \Delta x_i$.
 - Riemann sums are often partitioned into intervals of equal length, so let $\Delta x_i = (b-a)/n$ and each $c_i = a + \frac{b-a}{n}i$ for all i . The Riemann sum can be rewritten as: $\lim_{n \rightarrow \infty} \sum_{i=1}^n f\left(a + \frac{b-a}{n}i\right) \left(\frac{b-a}{n}\right)$. By recognizing this formula (rearranging the terms and determining a and b), we can reformulate a limit and summation problem as the definite integral $\int_a^b f(x) dx$, which is much easier to solve.
4. The First Fundamental Theorem of Integral Calculus: If $f(x)$ is continuous on (a, b) , then $\int_a^b f(x) dx = F(b) - F(a)$ for a function $F(x)$ such that $F'(x) = f(x)$ for all x .

Integration Rules and Theorems

- If $f(x)$ is continuous on $[a, b]$ then it is integrable on $[a, b]$. If g has a finite number of removable discontinuities on $[a, b]$ (and no other discontinuities), then $g(x)$ is also integrable on $[a, b]$.
- If f and g are integrable over $[a, b]$ and $f(x) \leq g(x)$ for all x in $[a, b]$, then $\int_a^b f(x) dx \leq \int_a^b g(x) dx$.
- If f is integrable over $[a, b]$ and $m \leq f(x) \leq M$ for all x in $[a, b]$, then $m(b-a) \leq \int_a^b f(x) dx \leq M(b-a)$.
- If $F(x)$ and $G(x)$ are antiderivatives of the same function, then $F(x) = G(x) + C$.
- Mean Value Theorem for Integration: If f is integrable on $[a, b]$, then there exists c such that $a \leq c \leq b$ and $f(c) = \frac{1}{b-a} \int_a^b f(x) dx$. The average value of f on $[a, b]$ is equal to $\frac{1}{b-a} \int_a^b f(x) dx$.
- Change of Variable in Definite Integrals: For $x = g(t)$, $a = g(t_1)$, and $b = g(t_2)$, then $\int_a^b f(x) dx = \int_{t_1}^{t_2} \left[f(g(t)) \cdot \frac{dg}{dt} \right] dt$.

7. Approximations of definite integrals

- a. Trapezoidal Rule: For $h = \frac{b-a}{n}$ and $x_i = a + ih$, then

$$\int_a^b f(x) dx \approx \frac{h}{2} \cdot (f(x_0) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-1}) + f(x_n)).$$

- b. Simpson's Rule: For an even integer n , $h = \frac{b-a}{n}$, and $x_i = a + ih$, then

$$\int_a^b f(x) dx \approx \frac{h}{3} \cdot (f(x_0) + 2f(x_1) + 4f(x_2) + 2f(x_3) + f(x_4) + \cdots + 2f(x_{n-1}) + f(x_n)).$$

Integration Formulas

$$\int c f(x) dx = c \int f(x) dx.$$

$$\int u^n du = \frac{u^{n+1}}{n+1} + C \quad (n \neq -1).$$

$$\int e^u du = e^u + C.$$

$$\int \cos u du = \sin u + C.$$

$$\int \tan u du = \ln|\sec u| + C = -\ln|\cos u| + C.$$

$$\int \sec^2 u du = \tan u + C.$$

$$\int \sec u \tan u du = \sec u + C.$$

$$\int \sec u du = \ln|\sec u + \tan u| + C.$$

$$\int \frac{1}{\sqrt{a^2 - u^2}} du = \arcsin \frac{u}{a} + C.$$

$$\int \frac{1}{\sqrt{u^2(a^2 - u^2)}} du = \frac{1}{a} \operatorname{arcsec} \frac{u}{a} + C.$$

$$\int [f(x) \pm g(x)] dx = \int f(x) dx \pm \int g(x) dx.$$

$$\int \frac{1}{u} du = \ln|u| + C.$$

$$\int a^u du = \frac{a^u}{\ln a} + C \quad (a > 0, a \neq 1).$$

$$\int \sin u du = -\cos u + C.$$

$$\int \cot u du = \ln|\sin u| + C = -\ln|\csc u| + C.$$

$$\int \csc^2 u du = -\cot u + C.$$

$$\int \csc u \cot u du = -\csc u + C.$$

$$\int \csc u du = -\ln|\csc u + \cot u| + C.$$

$$\int \frac{1}{a^2 + u^2} du = \frac{1}{a} \arctan \frac{u}{a} + C.$$

Techniques of integration

1. Substitution: Use of identities and change of variable.

2.* Integration by parts: $\int u dv = uv - \int v du$ or $\int_a^b u dv = uv|_a^b - \int_a^b v du.$

- 3.* Partial fractions: If the original denominator has a linear factor, place $ax+b$ in the new denominator with a constant numerator: $\frac{A}{ax+b}.$

4. For $\int \frac{p(x)}{q(x)} dx$, where the degree of $p(x)$ is greater than or equal to the degree of $q(x)$, divide the numerator by the denominator.

5. Completing the square: Use this technique for quadratic functions if necessary.

6. Splitting up fractions: $\int \frac{Ax+B}{ax^2+bx+c} dx = \int \frac{Ax}{ax^2+bx+c} dx + \int \frac{B}{ax^2+bx+c} dx.$

Applications of the Definite Integrals

1. Area:

- Area bounded by $y = f(x)$, x -axis, and lines $x = a$ and $x = b$ ($b > a$) equals $A = \int_a^b f(x) dx = \int_a^b y dx$.
- Area bounded by $x = g(y)$, x -axis, and lines $y = a$ and $y = b$ ($b > a$) equals $A = \int_a^b g(y) dy = \int_a^b x dy$.
- Since the area of a circle of radius a is πa^2 , $A = \int_0^a \sqrt{a^2 - u^2} du = \frac{1}{4} \pi a^2$.
- If $f(x) \geq g(x)$ on $[a, b]$, then the area bounded by the curves $y_1 = f(x)$, $y_2 = g(x)$, and the lines $x = a$ and $x = b$ is $A = \int_a^b (f(x) - g(x)) dx$.
- * Polar coordinates:

- The area bounded by $r = f(\theta)$ and two radii vectors $\theta = \alpha$ and $\theta = \beta$ is $A = \frac{1}{2} \int_{\alpha}^{\beta} r^2 d\theta$.
- The area between $r_1 = f(\theta)$ and $r_2 = g(\theta)$, from $\theta = \alpha$ and $\theta = \beta$ is $A = \frac{1}{2} \int_{\alpha}^{\beta} (r_1^2 - r_2^2) d\theta$.

2. Volume:

- Cross-sections: Volume of a solid whose cross-sectional area can be determined by $A(x)$ and whose cross sections are perpendicular to the x -axis is $A = \int_a^b A(x) dx$. (Similarly for y -axis)
- Solids of revolution:
 - Disk method:
 - Volume of solid generated by revolving region bounded by $y = R(x)$, $x = a$, and $x = b$ ($b > a$) about axis $y = c$ is $V = \pi \int_a^b [R(x) - c]^2 dx$.
 - Volume of solid generated by revolving region bounded by $x = R(y)$, $y = a$, and $y = b$ ($b > a$) about axis $x = c$ is $V = \pi \int_a^b [R(y) - c]^2 dy$.
 - Washer method:
 - Volume of solid generated by revolving region bounded by $y_1 = R(x)$, $y_2 = r(x)$, $x = a$, and $x = b$ ($b > a$) about axis $y = c$ is $V = \pi \int_a^b [(R(x) - c)^2 - (r(x) - c)^2] dx$.
 - Volume of solid generated by revolving region bounded by $x_1 = R(y)$, $x_2 = r(y)$, $y = a$, and $y = b$ ($b > a$) about axis $x = c$ is $V = \pi \int_a^b [(R(y) - c)^2 - (r(y) - c)^2] dy$.
 - Shell method:
 - Volume of solid generated by revolving region bounded by $y = f(x)$, x -axis, $x = a$, and $x = b$ ($b > a$) about axis $y = c$ is $V = 2\pi \int_a^b [(x - c)f(x)] dx$.
 - Volume of solid generated by revolving region bounded by $x = g(y)$, y -axis, $y = a$, and $y = b$ ($b > a$) about axis $x = c$ is $V = 2\pi \int_a^b [(y - c)g(y)] dy$.

3.* Arc length

- a. If $y = f(x)$ and $y' = f'(x)$ are continuous, then the arc length of $y = f(x)$ from $x = a$ to $x = b$ ($b > a$) is $s = \int_a^b \sqrt{1 + y'^2} dx$. Similarly, for $x = g(y)$ and $x' = g'(y)$, from $y = a$ to $y = b$ ($b > a$), $s = \int_a^b \sqrt{1 + x'^2} dy$.
- b. Parametrically, if $x = f(t)$ and $y = g(t)$, then the arc length from t_1 to t_2 is $s = \int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$.
- c. In polar coordinates, the arc length of the curve $r = f(\theta)$ from $\theta = \alpha$ to $\theta = \beta$ is $s = \int_{\alpha}^{\beta} \sqrt{r^2 + r'^2} d\theta$.

4.* Work: Motion Along a Straight Line

- a. $W = fd$ (for a constant force only).
- b. Total work done in moving an object along the x -axis from $x = a$ to $x = b$, if $f(x)$ is a force exerted at a point x , is $W = \int_a^b f(x) dx$.
- c. Hooke's Law: $f \propto x$ or $f = kx$ (for spring constant k and displacement x). Therefore, the work done in compressing a spring from point $x = a$ to $x = b$ is $W = \int_a^b kx dx$.

5.* Improper integrals

- a. Convergent: approaches a limiting value. Divergent: Approaches no value (∞ , oscillates, etc.).
- b. Discontinuous (non-removable) integrand
- Integrand becomes ∞ at lower limit: $\int_a^b f(x) dx = \lim_{h \rightarrow 0^+} \int_{a+h}^b f(x) dx$.
 - Integrand becomes ∞ at upper limit: $\int_a^b f(x) dx = \lim_{h \rightarrow 0^+} \int_a^{b-h} f(x) dx$.
 - Integrand becomes ∞ at point c in $[a, b]$: $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$, and use i and ii.
- c. Infinite limits of integration.
- $\int_a^\infty f(x) dx = \lim_{b \rightarrow \infty} \int_a^b f(x) dx$.
 - $\int_{-\infty}^b f(x) dx = \lim_{a \rightarrow -\infty} \int_a^b f(x) dx$.
 - $\int_{-\infty}^\infty f(x) dx = \int_{-\infty}^c f(x) dx + \int_c^\infty f(x) dx$ (for suitable c).

*Sequences and Series

1. Definitions

- a. Def'n: A sequence is a function whose domain is restricted to the set of integers. The values in the range of the sequences are its elements or terms.
- b. Def'n: A sequence $\{a_n\}$ has the limit L , $\lim_{n \rightarrow \infty} a_n = L$, if for all $\varepsilon > 0$ there exists an integer $N > 0$ such for all $n \geq N$, $0 \leq |a_n - L| < \varepsilon$.
- c. Def'n: A sequence that has a finite limit is said to converge, while sequences that do not have a finite limit are said to diverge.

2. Basic theorems

- a. If $\lim_{x \rightarrow \infty} f(x) = L$ and f is defined for all positive integers, then also $\lim_{n \rightarrow \infty} f(n) = L$, where n is a positive integer.

- b. The limit of a convergent sequence is unique.
 c. If $\{a_n\}$ and $\{b_n\}$ are convergent sequences and C is a constant, then

i. $\lim_{n \rightarrow \infty} C a_n = C \lim_{n \rightarrow \infty} a_n$.
 ii. $\lim_{n \rightarrow \infty} (a_n \pm b_n) = \lim_{n \rightarrow \infty} a_n \pm \lim_{n \rightarrow \infty} b_n$.
 iii. $\lim_{n \rightarrow \infty} a_n b_n = \lim_{n \rightarrow \infty} a_n \cdot \lim_{n \rightarrow \infty} b_n$.
 iv. $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{\lim_{n \rightarrow \infty} a_n}{\lim_{n \rightarrow \infty} b_n}$ (if $\lim_{n \rightarrow \infty} b_n \neq 0$).

3. Monotonic sequences

- a. Def'n: A sequence is said to be monotonic and
 i. increasing (or non-decreasing) if $a_n \leq a_{n+1}$ for all n .
 ii. strictly increasing if $a_n < a_{n+1}$ for all n .
 iii. decreasing (or non-increasing) if $a_n \geq a_{n+1}$ for all n .
 iv. strictly decreasing if $a_n > a_{n+1}$ for all n .

4. Bounds

- a. Definitions:
 i. A is the upper bound of the sequence $\{a_n\}$ if $a_n \leq A$ for all positive integers n .
 ii. B is the lower bound of the sequence $\{a_n\}$ if $a_n \geq B$ for all positive integers n .
 iii. A sequence is bounded if there exists C such that $|a_n| \leq C$ for all positive integers n (i.e., the sequence is bounded from above and below).
 b. Theorem: A monotonic sequence is convergent if and only if it is bounded.

*Infinite Series

1. Definitions and Basic Theorems

- a. Def'n: If $\{a_n\}$ is a sequence, then an expression of the form $\sum_{n=1}^{\infty} a_n = a_1 + a_2 + \cdots + a_m + \cdots$ is called an infinite series.
 b. Def'n: For infinite series $\sum_{n=1}^{\infty} a_n$ and sequence $\{S_n\}$, $S_n = \sum_{n=1}^n a_n = a_1 + a_2 + \cdots + a_n$ is called the partial sum of the series. If $\lim_{n \rightarrow \infty} S_n$ exists and is equal to S , the infinite series $\sum_{n=1}^{\infty} a_n$ is convergent and S is the sum of the series ($S = \sum_{n=1}^{\infty} a_n$). If $\lim_{n \rightarrow \infty} S_n$ does not exist, then the series is divergent and has no sum.
 c. If the infinite series $\sum_{n=1}^{\infty} a_n$ is convergent, then $\lim_{n \rightarrow \infty} a_n = 0$. (Note: The converse is not true).
 Corollary: If $\lim_{n \rightarrow \infty} a_n \neq 0$, then the series is divergent.
 d. For any positive integer k , $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=k}^{\infty} a_n$ both either converge or diverge.
 e. If c is a non-zero constant and $\sum a_n$ converges, then the series $\sum c a_n$ converges, and its limit is equal to $c \sum a_n$. If $\sum a_n$ diverges, then $c \sum a_n$ diverges as well.

- f. If Σa_n and Σb_n are both convergent, then $\Sigma(a_n \pm b_n)$ is also convergent and $\Sigma(a_n \pm b_n) = \Sigma a_n \pm \Sigma b_n$.
2. Some Important Series
- a. Geometric Series: Σar^n is convergent if $|r| < 1$ (otherwise it diverges), and its sum equals $a/(1-r)$.
- b. P-series (Hyperharmonic): For positive integer p , $\Sigma \frac{1}{n^p}$ converges only for $p > 1$.
- Note: The harmonic series $\Sigma \frac{1}{n}$ is simply a p-series with $p = 1$. The harmonic series diverges.
- c. The series $\Sigma \frac{1}{n!}$ is convergent (the sum of the series is e , the base of the natural logarithm).
- d. Polynomial Test: If $P(n)$ and $Q(n)$ are polynomials of degree p and q respectively, then $\Sigma \frac{P(n)}{Q(n)}$ is convergent if $q > p+1$ and diverges if $q \leq p+1$.
3. Alternating series and absolute convergence
- a. Def'n: An infinite series of the form $\sum_{k=1}^{\infty} (-1)^{k+1} a_k = a_1 - a_2 + a_3 - a_4 + \dots$, where $a_k > 0$ for all k , is called an alternating series.
- b. Alternating Series Test: The alternating series $\sum_{k=1}^{\infty} (-1)^{k+1} a_k$ converges if $0 < a_{n+1} < a_n$ for all n and $\lim_{n \rightarrow \infty} a_n = 0$.
- c. For the alternating series $\sum_{k=1}^{\infty} (-1)^{k+1} a_k$, the error made by using the first k terms as an approximation for the sum of the series is less than a_{k+1} .
- d. Absolute Convergence
- Def'n: The infinite series Σa_n converges absolutely if $\Sigma |a_n|$ converges.
 - Def'n: If a series is convergent but not absolutely convergent, then it is said to be conditionally convergent.
 - If a series converges absolutely, then it also converges.
4. Convergence Tests
- Def'n: A series of only positive terms is a positive series.
 - Theorem: A necessary and sufficient condition for an infinite positive series to be convergent is that its sequence of partial sums has an upper bound.
 - Direct Comparison test: For positive series Σa_n and Σb_n ,
 - If Σb_n is convergent and $a_n \leq b_n$ for all n , then Σa_n is also convergent.
 - If Σb_n is divergent and $a_n \geq b_n$ for all n , then Σa_n is also divergent.
 - Integral Test: If f is positive, continuous, and decreasing for all $x \geq 1$ and $a_n = f(n)$, then $\sum_{n=1}^{\infty} a_n$ and $\int_1^{\infty} f(x)dx$ either both converge or both diverge. If the series converges to a sum S , then the error after the n th term approximation, $S - S_n$, is given by $0 \leq R_n \leq \int_n^{\infty} f(x)dx$.
 - Ratio Test: For a positive series Σa_n with nonzero terms, let $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L$
 - If $L < 1$, the series converges.

- b. If $L > 1$, the series diverges.
 - c. If $L = 1$, the test fails.
6. Limit Comparison Test: For positive series $\sum a_n$ and $\sum b_n$, if $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} > 0$, then $\sum a_n$ and $\sum b_n$ are either both convergent or both divergent.
7. Root Test: For positive series $\sum a_n$, let $\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = L$
- a. If $L < 1$, the series converges.
 - b. If $L > 1$, the series diverges.
 - c. If $L = 1$, the test fails.

*Power Series

1. Definitions and Basic Theorems

- a. Def'n: If $\{a_n\}$ is a sequence of constants, the expression $\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n$ is called a power series in x .
- b. If the power series $\sum a_n x^n$ converges for a number $c \neq 0$, then it converges absolutely for all x such that $|x| < |c|$.
- c. If the power series $\sum a_n x^n$ diverges for a number c , then it diverges for all x such that $|x| > |c|$.
- d. Let $\sum a_n x^n$ be a given power series in x . Then exactly one of the following holds:
 - i. The series converges only when $x = 0$.
 - ii. The series converges for all values of x .
 - iii. There exists a number $r > 0$ (called the radius of convergence) such that the series is absolutely convergent for all values of x for which $|x| < r$ and is divergent for all values of x for which $|x| > r$ (the series may converge at one of both of the endpoints r and $-r$).
- e. Def'n: The set of all values for which a power series is convergent is called the interval of convergence of the power series.
- f. Def'n: Similarly, if a_n is a sequence, we define the expression $\sum_{n=0}^{\infty} a_n (x-b)^n = a_0 + a_1 (x-b) + a_2 (x-b)^2 + a_3 (x-b)^3 + \dots$ to be a power series in $(x-b)$.
- g. Similarly, for the power series $\sum a_n (x-c)^n$ in $(x-c)$,
 - i. The series converges only when $x = c$.
 - ii. The series converges for all values of x .
 - iii. There exists a number $r > 0$ (called the radius of convergence) such that the series is absolutely convergent for all x such that $|x - c| < r$ and is divergent for all values of x such that $|x - c| > r$ (the series may converge at one of both of the endpoints $c + r$ and $c - r$).

2. More on Power Series

- a. Def'n: Since the power series $\sum a_n (x-b)^n$ has a unique sum at each point in its interval of convergence, a function f is defined by the power series, i.e. $f(x) = \sum a_n (x-b)^n$. The domain of f is the interval of convergence of the power series and the values of f at any point x in this domain is the sum of the power series at this point.
- b. Def'n: The derived series of the power series $\sum a_n (x-b)^n$ is $\sum n a_n (x-b)^{n-1} = a_1 + 2a_2(x-b) + 3a_3(x-b)^2 + \dots + n a_n (x-b)^{n-1} + \dots$, i.e., the series formed by taking the derivatives of the terms of the original series.

c. Functions defined by power series behave very much like polynomials as is indicated by the following theorem. Thm: Let $\sum a_n(x-b)^n$ be a power series in $(x-b)$ and let f be defined by $f(x) = \sum a_n(x-b)^n$, where x is any number in the interval of convergence of the series.

- i. f is continuous at every point in the interval of convergence.
- ii. The derived series of the power series is equal to the derivative of the function defined by the power series at every point in the interval of convergence, except possibly at the endpoints, i.e., $f(x) = \sum D_x[a_n(x-b)^n] = \sum na_n(x-b)^{n-1}$.

iii. The series obtained by integrating the terms of the given series converge to $\int_b^x f(t)dt$ for each x within the interval of convergence of the original power series, except possibly at the

endpoints, i.e. $\int_b^x f(t)dt = \sum_{n=0}^{\infty} \int_b^x a_n(t-b)^n dt = \sum_{n=0}^{\infty} a_n \frac{(x-b)^{n+1}}{n+1} = a_0(x-b) + a_1(x-b)^2 + \dots$

*Taylor Series and Taylor's Theorem

1. If a function f is infinitely differentiable, the series $\sum_{n=0}^{\infty} \frac{f^{(n)}(b)(x-b)^n}{n!} = f(b) + f'(b)(x-b) + \frac{f''(b)(x-b)^2}{2!} + \frac{f'''(b)(x-b)^3}{3!} + \dots$ is called the Taylor series in $(x-b)$ of the function.

2. Consider any power series in $(x-b)$, $a_0 + a_1(x-b) + a_2(x-b)^2 + \dots$ and denote its interval of convergence by $(b-r, b+r)$, where $0 \leq r < \infty$. Denote by the function defined as $f(x) = a_0 + a_1(x-b) + a_2(x-b)^2 + \dots + a_n(x-b)^n + \dots$ where x is any number in $(b-r, b+r)$. Then the power series is

the Taylor's series in $(x-b)$ of the function f , namely $f(x) = f(b) + f'(b)(x-b) + \frac{f''(b)(x-b)^2}{2!}$

$+ \frac{f'''(b)(x-b)^3}{3!} + \dots + \frac{f^{(n)}(b)(x-b)^n}{n!} + \dots$ A Taylor series in $(x-b)$ of a function f is sometimes called the Taylor expansion of f about the point b .

3. The Taylor series in x is called a Maclaurin series, $f(0) + f'(0)x + \frac{f''(0)x^2}{2!} + \frac{f'''(0)x^3}{3!} + \dots$

$+ \frac{f^{(n)}(0)x^n}{n!} + \dots$ A Taylor series in $(x-b)$ of a function f is sometimes called the Taylor expansion of f about the point b .

4. Taylor's Theorem: Let f be a function which is defined in some open interval $(b-r, b+r)$, $0 \leq r < \infty$, and whose first $n+1$ derivatives exist and are continuous throughout the interval. Then for all x

in $(b-r, b+r)$, $f(x) = f(b) + f'(b)(x-b) + \frac{f''(b)(x-b)^2}{2!} + \dots + \frac{f^{(n)}(b)(x-b)^n}{n!} + R_n(x)$ where

$$R_n(x) = \frac{f^{(n+1)}(\xi)(x-b)^{(n+1)}}{(n+1)!} \quad (\text{Lagrange form of the remainder}) \text{ and } b < \xi < x.$$

***Important Series Expansions**

1. $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots + \frac{(-1)^{n-1} x^{2n-1}}{(2n-1)!} + \dots \quad (-\infty < x < \infty).$
2. $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots + \frac{(-1)^{n-1} x^{2n-2}}{(2n-2)!} + \dots \quad (-\infty < x < \infty).$
3. $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!} + \dots \quad (-\infty < x < \infty).$
4. $\ln x = (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \frac{(x-1)^4}{4} + \dots + \frac{(-1)^{n-1} (x-1)^n}{n} + \dots \quad (0 < x \leq 2).$
5. $\frac{1}{x} = 1 - (x-1) + (x-1)^2 - (x-1)^3 + \dots + (-1)^n (x-1)^n + \dots \quad (0 < x < 2).$
6. $\frac{1}{1+x} = 1 - x + x^2 - x^3 + \dots + (-1)^n x^n + \dots \quad (-1 < x < 1).$
7. $\frac{1}{1+x^2} = 1 - x^2 + x^4 - x^6 + \dots + (-1)^n x^{2n} + \dots \quad (-1 < x < 1).$
8. $\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots + \frac{(-1)^{n-1} x^{2n-1}}{2n-1} + \dots \quad (-1 \leq x \leq 1).$
9. The General Binomial Theorem: $(1+x)^m = 1 + mx + \frac{m(m-1)}{2!} x^2 + \dots + \frac{m(m-1)\dots(m-n+1)}{n!} x^n + \dots$
(for all real m , $-1 < x < 1$).

Best wishes to
Math Survey
from the
Stuyvesant H.S.
Parents' Association.

Math Survey

would like to give
special thanks to the
Alumni Association
for its generous support.

Math Survey would like to thank our benefactors and sponsors.

Gianni Babino
Eric Chen
John Choi
Edmund Chou
Jean Chou
Naomi Davis
Ruth Davis

Soon Sun Eum
Rudy Huang
Hyoonie
Brandon Irizarry
Paul "SUPERMATH" Kim
Sung Hoon Kim
Weikai Lang

Ms. Latham
Ms. Latham
Ms. Latham
Shizuko Mikurya
David Tischler
Maria Uhan
Nelson Uhan

S. Adnolf-Ginnerstein
Sitij Agrawal
Anonymous
Jesse Antila-Hughes
Jesse Antila-Hughes
David Bosko
Amelia Bucek
Huiwen Chan
Monica Chan
James Chen
Monte Chin
Jane Cho
Amy Chou
Wayne Chung
Erin Donar
Anna Douglas
Robert Edele
Annie Eisenberg
Nick Fayco
Sophia Feng
Tal Fishman
William Friedman
Helena Moby Fu
Edward Fung
Elizabeth Garcia
Elaine Gee
Mr. Geller
Aaron Goldsmid
Joshua Goldwitz
Scott Gordon
Salerie Gorodetskaya
Boris Granovskiy
Sharmin Haque
Xining He

William Ho
Fraustina Hsu
Victoria Hsu
William Hsu
Randy Ip
IRS
David Jang
Jae Jang
Micheal Jha
Graham Johnson
Zardosht Kasheff
Nora Kelleher
Shohye Kim
Tae Jin Kim
Eddie Kotlyar
Christina Ku
Vy Lam
Wilstak Lam
Weikai Lang
Weikai Lang
Tim Lau
Jenny Lee
Joe Lee
Kathernie Lee
Roger Lee
Sujean Lee
Brian Leung
Jennifer Leung
Victor Lew
Eric Li
Anita Liao
Charles Lim
Edward Lin
Shirley Lin

Tony Lin
Michael Lipatov
Aimee Liu
Lian Liu
Timothy Louie
Rona Luo
Vadim Lyubashevsky
Vadim Lyubashevsky
Josh Malamy
Kunihiko Mikurya
Kunihiko Mikurya
Yun Ki Min
Fongo Mongo
Deborah Morns
Antonie Munfakh
Elan Naveh
Igoz Nepohinyathchiy
Howard Ng
Linda Ng
Natan Ovshy
Sandy Pae
Palak Parekh
Palak Parekh
Parthipan Pathmanapan
Ting Pen
Aaron Perlstein
Fan Ping
Jack Po
Namita Puritah
Graham Sack
Paul Santini
Nancy Schkvga
Polly Sepowitz
Farhan Shamst

Idan Shani
Austin Shau
Bo Shen
S. Shiffman-Ackerman
Alla Sludsker
Bijal Soni
Edward Sun
Edward Sun
Edward Sun
Jia Sun
Elaine Tan
Alex Tang
Alex Tang
Samantha Trray
Linn Tynose
Noel Vega
Andrew Villicastin
Charles Wang
David Wang
Marc Weiskoff
Joseph Weiss
Kirstin Wickelgren
Christina Won
Gary Wong
Irene Wong
Julia Wu
Lisa Wu
Rathin Yaguik
Amy Yang
Annie Yao
Denis Yotama
Nancy Zam
Chou Zhao
Juikhail Zheurorsky